

Amazon Dangerous Goods Classification Upload File

Rechargeable Lithium Ion Battery - BL1860B 18V 6Ah 108Wh

ASIN	B0GS9F72TY
Package Quantity	6 Pack
Product Title	6 Pack 6Ah 18V Lithium Battery Replacement Compatible with Makita LXT Batteries BL1860B BL1850B BL1840B BL1830B BL1820 BL1815, Rechargeable Battery Pack for Cordless Power Tools
Battery Model	BL1860B
Rating	18.0V, 6.0Ah, 108Wh
Battery Type	Rechargeable Lithium Ion Battery
Manufacturer	Guangdong Alfdeco Smart Energy Co., Ltd

Attached supporting documents in this PDF:

- MSDS / SDS Report: YL251145001M01
- UN38.3 Test Summary: YL251145001U01
- UN38.3 Test Report: YL251145001U01
- IEC 62133-2 Safety Test Report: YL251145002

Note: This cover page is prepared to identify the ASIN for Amazon review. The compliance documents attached after this page apply to the BL1860B 18V 6Ah 108Wh rechargeable lithium-ion battery model used by this listing.

Amazon Dangerous Goods / Lithium Battery Compliance Package

For Product Classification Review - Model BL1860B

Product Name	Rechargeable Lithium Ion Battery
Model	BL1860B
Rating	18.0V, 6.0Ah, 108Wh
Battery Type	Lithium-ion rechargeable battery pack
Manufacturer	Guangdong Alfdeco Smart Energy Co., Ltd
Manufacturer Address	Room 501, Building 2, No.7, Luoqunpu 1st Road, Qingxi Town, Dongguan City, Guangdong Province

Included Documents

No.	Document	Report / Reference No.	Issue Date / Key Result
1	Material Safety Data Sheet (MSDS)	YL251145001M01	2025-11-06
2	UN38.3 Test Summary	YL251145001U01 / PEKSZ202511119665	T.1-T.8: Pass
3	UN38.3 Test Report	YL251145001U01	2025-11-06; UN Manual of Tests and Criteria Rev.8, Section 38.3
4	IEC 62133-2 Safety Test Report	YL251145002	2025-11-06; Product fulfils EN 62133-2:2017/A1:2021

Notes for Amazon Review

- This package is intended for Amazon dangerous goods / lithium battery classification review.
- The product is a lithium-ion rechargeable battery pack: 18V, 6Ah, 108Wh.
- The UN38.3 summary and test report show required transport tests T.1 to T.8 with passing results.
- SDS/MSDS is included for hazard and transport handling information.
- CE, EMC, RoHS and EU Battery Regulation documents are not included here because they are usually used for separate product compliance requests rather than DG classification.

MSDS Report

Applicant: 申请商:	Guangdong Alfdeco Smart Energy Co., Ltd 广东艾弗得智慧能源有限公司 Room 501, Building 2, No.7, Luoqunpu 1st Road, Qingxi Town, Dongguan City, Guangdong Province 广东省东莞市清溪镇罗裙埔一路 7 号 2 号楼 501 室
Manufacturer: 制造商:	Guangdong Alfdeco Smart Energy Co., Ltd 广东艾弗得智慧能源有限公司 Room 501, Building 2, No.7, Luoqunpu 1st Road, Qingxi Town, Dongguan City, Guangdong Province 广东省东莞市清溪镇罗裙埔一路 7 号 2 号楼 501 室
Product Name: 产品名称:	RECHARGEABLE LITHIUM ION BATTERY 可充电锂离子电池组
Model 型号:	BL1860B
Inspection according to 鉴定依据	UN "Recommendations on the TRANSPORT OF DANGEROUS GOODS" 联合国《关于危险品货物运输的建议书》
Prepared By : 编制单位:	Shenzhen Yilink Testing Co., LTD. 深圳亿联检测有限公司 Room 1406, Building A, Meixun Science Park, No.19 Jinxiu Middle Road, Laokeng Community, Longtian Street, Pingshan District, Shenzhen 深圳市坪山区龙田街道老坑社区锦绣中路 19 号美讯数码科技厂区 1 号厂房 1406
Report No. 报告编号:	YL251145001M01

Written by 编写: Jack Tang

Inspected by 审核: Jack Tang

Approved by 批准: Noah shaw

Date 日期: 2025.11.06



Material Safety Data Sheet

化学品安全技术说明书

1. Identification of the product and supplier (产品和厂商信息)	
Product Name: 产品名称:	RECHARGEABLE LITHIUM ION BATTERY 可充电锂离子电池组
Model: 型号:	BL1860B
Rating: 规格:	18V 6Ah 108Wh
Battery Size: 电池尺寸:	(117.0 x 75.0 x 67.0) mm
Weight: 重量:	570.7g
Factory: 生产单位:	Guangdong Alfdeco Smart Energy Co., Ltd. 广东艾弗得智慧能源有限公司
Factory address: 生产单位地址:	Room 501, Building 2, No.7, Luoqunpu 1st Road, Qingxi Town, Dongguan City, Guangdong Province 广东省东莞市清溪镇罗裙埔一路7号2号楼501室
Emergency telephone call: 紧急联系电话:	0755-84571435
E-mail: 邮箱:	guobinsheng@jingnoo.com

2. Composition/Information on Ingredients (成分/组成信息)

Chemical Name 化学名称	CAS Number CAS 号 (化学文摘索引登记号)	Concentration or concentration ranges (%) 浓度或浓度范围(%)
Lithium Nickel Cobalt Manganate 镍钴锰酸锂(Li(NiCoMn)O ₂)	182442-95-1	25~35
Graphite 石墨 (C)	7782-42-5	15~20
Polyvinylidene Fluoride 聚偏氟乙烯(PVDF)	24937-79-9	1~5
Acetylene Black 乙炔黑(SP)	1333-86-4	0.5~3
Copper 铜(Cu)	7429-90-5	21~23
Lithium hexafluorophosphate 六氟磷酸锂(LiPF ₆)	7440-50-8	10~11

N/A=Not applicable.

N/A 表示不适用。

3. Hazards Identification (危险性概述)

Hazard Description 危险性描述	Not dangerous with normal use. Do not dismantle, open or shred the battery ingredients contained within or their ingredients products could be harmful. 正常使用没有危险, 不能拆解、打开或分解电池, 里面的材料或成分是有危害的。
Primary Route(s) of Exposure 接触途径	Inhalation, Ingestion, Skin contact and Eye contact. 吸入、食入、皮肤接触、眼睛接触。
Potential Health Effects 潜在健康影响	Inhalation: Vapors or mists from a ruptured battery may cause respiratory irritation. 吸入: 破裂的电池散发出来的气雾会引起呼吸道刺激。 Ingestion: The battery ingredients contained within or their ingredients products can cause serious chemical burns of mouth, esophagus, and gastrointestinal tract. 食入: 电池的组成成分或原料可以导致嘴, 食道和胃肠道的严重化学烧伤。 Skin: Skin contact with contents of an open battery can cause severe irritation or burns to the skin. 皮肤: 皮肤接触到电池的內部化学材料可能会导致严重的刺激或烧伤皮肤。 Eye: Eye contact with contents of an open battery can cause severe irritation or burns to the eye.

眼睛: 眼睛接触到电池的內部化学材料可能会导致严重的刺激或烧伤眼睛。

4. First Aid Measures (急救措施)

Inhalation 吸入	Remove source of contamination or move victim to fresh air. Obtain medical advice. 移除污染源或者将受害者移至新鲜空气处。寻求医生建议。
Ingestion 食入	Please rinse mouth thoroughly with water. Induce vomiting under the guidance of professional personage. Please seek medical treatment in time. 立即用清水漱口，在专业人士的指导下催吐，速就医。
Skin contact 皮肤接触	Remove contaminated clothes and rinse skin with plenty of water or shower for 15 minutes. Get medical aid. 脱下已污染衣服，用大量的水冲洗至少 15 分钟，速就医。
Eye contact 眼睛接触	Irrigate with flowing water for 15 minutes. If irritation persists, consult a physician. 用流动水冲洗 15 分钟，如刺激持续发生，请求助于医生。

5. Fire Fighting Measures (消防措施)

Characteristics of Hazard 危险特性	Toxic fumes, gases or vapors may evolve on burning. 火灾时可释放有害浓烟、气体或者蒸汽。
Hazardous Combustion Products 燃烧产生的危险物品	Carbon monoxide, carbon dioxide, lithium oxide fumes and so on. 一氧化碳，二氧化碳，锂氧化物烟气等。
Fire-extinguishing Methods and Extinguishing Media 灭火方法及灭火剂	Please use water, dry sand and other proper fire extinguishing media. 请使用水，干燥沙等合适的灭火介质。
Attention in Fire-extinguishing 灭火注意事项	The firemen should put on antigas masks and full fire-fighting suits. 消防人员须佩戴防毒面具、穿全身消防服。

6. Accidental Release Measures (泄露应急处理)

Personal Precautions, protective equipment, and emergency procedures 个人预防措施、防护装备和应急程序	Restrict access to area until completion of clean-up. Do not touch the spilled material. Wear adequate personal protective equipment as indicated in Section 8. 限制区域，直到完成清理工作。请勿触摸泄漏的材料。穿戴适当的个人防护设备，如第 8 部分所示。
Environmental Precautions 环境保护措施	Prevent material from contaminating soil and from entering sewers or waterways. 防止物质污染土壤和进入下水道或水道。
Methods and materials for cleaning up 清理的方法和材料	Absorb spilled material with an inert absorbent (dry sand or earth). Scoop contaminated absorbent into an acceptable waste container. Collect all contaminated absorbent and dispose of according to directions in Section 13. Scrub the area with detergent and water; collect all contaminated wash water for proper disposal. 用惰性吸收剂(干砂或沙土)吸收溢出的材料。污染物转移到可吸收废物的容器。收集所有受污染的吸收剂和根据第 13 部分的指令处置。用洗涤剂和水清洁污染区域，收集所有受污染的洗涤水进行适当处置。

7. Handling and Storage (操作处置与储存)

Handling 操作	Don't handling the batteries in manner that allows terminals to short circuit. Do not open, disassemble, crush or burn battery. 不要以让接头短路的方式对电池进行操作。不要打开，分解，挤压或燃烧电池。
Storage 储存	If the battery is subject to storage for such a long term as more than 3 months, it is recommended to recharge the battery periodically. 如果电池长期存放超过 3 个月，建议定期对电池充电。 Long period storage: -10℃~35℃, 60±25%R.H 长期存储: -10℃~35℃, 相对湿度 60±25% Do not storage the battery haphazardly in a box or drawer where they may short-circuit each other or be short-circuited by other metal objects. 不要将电池随意丢在盒子或抽屉里，以免电池之间或电池与其他金属物质发生短路。 Keep out of reach of children. 储存在小孩接触不到的地方。 Do not expose the battery to heat or fire. Avoid storage in direct sunlight. 不要将电池暴露在火源和热源附近，避免在阳光直射下存储。 Do not store together with oxidizing and acidic materials. 不要与氧化和酸性物质存储在一起。

8. Exposure Controls/Personal Protection (接触控制和个体防护)

Engineering Controls 工程控制	No engineering controls are required for handling batteries that have not been damaged. Personal protective equipments for damaged batteries should include chemical resistant gloves and safety glasses. 操作未破损的电池，没有工程控制要求。对于破损的电池，个人防护用品应包括化学品防护手套和安全眼镜。
Personal Protective Equipment 个人防护设备	Respiratory Protection: In case of battery venting, provide as much ventilation as possible. Avoid confined areas with venting cell cores. Respiratory Protection is not necessary under conditions of normal use. Not necessary under conditions of normal use. 呼吸保护：当电池排气阀打开时，应尽量使通风设备开至最大，避免将打开排气阀的电芯局限在某一狭窄空间内。正常操作条件下，呼吸保护是不必要的。正常使用条件下不必考虑。 Protective Gloves: Not necessary under conditions of normal use. 防护手套：正常使用条件下不必考虑。 Other Protective Clothing or Equipment: Not necessary under conditions of normal use. 其他防护服装或设备：正常使用条件下不必考虑。 Personal Protection is recommended for venting battery: Respiratory Protection, Protective Gloves, Protective Clothing and safety glass with side shields. 当电池排气阀打开时，应做好个人防护：呼吸防护，防护手套，防护服装和有护边的安全玻璃罩都是要准备的。

9. Physical and Chemical Properties (物理和化学特性)

Physical State 物理状态	Form: Solid 形态：固体
	colour : Black and cyan 颜色：黑色和青色
	Odour: Odorless 气味：无气味
pH 酸碱度	Not applicable as supplied 不适用
Flash Point 燃点	Not applicable unless individual components exposed 除单个电芯暴露试验外其他不适用
Flammability 可燃性	Not applicable unless individual components exposed 除单个电芯暴露试验外其他不适用
Relative density 相对密度	Not applicable unless individual components exposed 除单个电芯暴露试验外其他不适用
Solubility (water)	Not applicable unless individual components exposed

溶解性（水溶性）	除单个电芯暴露试验外其他不适用
Solubility (other) 溶解性（其他）	Not applicable unless individual components exposed 除单个电芯暴露试验外其他不适用
Vapor Density: (Air = 1) 蒸汽密度: (空气= 1)	Not applicable unless individual components exposed 除单个电芯暴露试验外其他不适用
Density/relative density 密度/相对密度	Not applicable unless individual components exposed 除单个电芯暴露试验外其他不适用
Solubility in Water 水溶性	Not applicable unless individual components exposed 除单个电芯暴露试验外其他不适用

10. Stability and Reactivity（稳定性和反应性）

Stability 稳定性	Stable under normal temperatures and pressures. 常温常压下稳定。
Conditions to Avoid 应避免的条件	Heat above 70° C or Incinerate, Deform, Mutilate, Crush, Disassemble, Overcharge, Short circuit, Expose over a long period to humid conditions. 加热 70° C 以上或焚烧、变形、毁坏、粉碎、拆卸、过充电、短路，长时间暴露在潮湿的条件下。
Hazardous Decomposition Products 危害分解物	Toxic Fumes, and may form peroxides. 有毒烟雾，并可能形成过氧化物。
Possibility of Hazardous Reaction 危险反应的可能性	If leaked, forbidden to contact with strong oxidizers, mineral acids, strong alkalis, halogenated hydrocarbons. 如果发生泄露，避免与强氧化剂，无机酸，强碱，卤代烃接触。

11. Toxicological Information（毒理学信息）

Irritation 刺激	In the event of exposure to internal contents, vapor fumes may be very irritating to the eyes and skin. 内部物质暴露的情况下，蒸汽烟雾可能对眼睛和皮肤产生刺激性。
Sensitization 致敏	No data is available 无数据可提供
Reproductive Toxicity 再生毒性	No data is available 无数据可提供
Toxicologically Synergistic Materials 协同材料毒理学	No data is available 无数据可提供

12. Ecological Information (生态学信息)

General note 通用信息	Do not allow undiluted product or large quantities of it to reach ground water, water course or sewage system. 不允许未稀释或大量的产品到达地下水、水道或污水系统。
Anticipated behavior of a chemical product in environment/possible environmental impact/ecotoxicity 化学产品在环境/可能的环境预期的行为的一种生态毒性	No data is available 无数据可提供
Mobility in soil 土壤中移动性	No data is available 无数据可提供
Persistence and Degradability 持久性和降解性	No data is available 无数据可提供

13. Disposal Considerations (废弃处置)

Waste Treatment 废弃处置方法	Recycle or dispose of in accordance with government, state & local regulations. 建议遵照国家和地方法规处置或再利用。
Attention for Waste Treatment 废弃注意事项	Deserted batteries couldn't be treated as ordinary trash. Couldn't be thrown into fire or placed in high temperature. Couldn't be dissected, pierced, crushed or treated similarly. Best way is recycling. 废电池不能被当做普通垃圾。不能扔进火中或置于高温下。不能解体, 刺穿, 破碎或类似的处理。最好的办法是回收利用。

14. Transport Information (运输信息)

Label for conveyance 运输标签	Lithium Battery Mark 锂电池操作标签	
UN Number UN 编号	UN3480 or UN3481 UN3480 或 UN3481	
Transport hazard class(es) 运输风险类别	Class 9 9 类危险品	
Packing Group 包装等级	965 or 966	II
	967	--
EmS No EmS 编号	F-A, S-I	
Marine pollutant(Y/N) 海洋污染物 (Y / N)	N	
Proper Shipping name 正确的装运名称	1) Lithium ion batteries; 2) Lithium ion batteries packed with equipment; 3) Lithium ion batteries contained in equipment. (including Lithium ion polymer batteries) 1) 锂离子电池; 2) 锂离子电池伴随设备包装在一起 3) 锂离子电池装在设备中 (包括锂离子聚合物电池)	
Hazard Classification 危险分类	The goods shall be complied with the requirements of Section IA of 965 or Section I of 966~967 of 66 th DGR Manual of IATA (2025 Edition) and Special Provision 188 of IMDG CODE (Amdt. 42-24), including the passing of the UN38.3 test. 货物应遵守 IATA 第 66 版 DGR 手册包装说明 965 第 IA 节规定或 966~967 第 I 节规定(2025 年版), 和特殊规定 188 海运危险货物规则(Amdt. 42-24), 包括通过 UN38.3 测试手册要求。	
ADR/ADN: 欧洲国际陆运危险货物协定/关于内陆水道国际运输危险货物的欧洲协定:	Transport Requirements for United Nations Economic Commission for Europe (UNECE) ADR/ADN, Applicable as from 1 January 2023. 自 2023 年 1 月 1 日起需符合联合国欧洲经济委员会(UNECE)ADR /ADN 的运输要求。	

15. Regulatory Information (法规信息)

《Dangerous Goods Regulations》
 《危险物品规则》
 《Recommendations on the Transport of Dangerous Goods Model Regulations》
 《危险货物运输的建议模型规定》
 《International Maritime Dangerous Goods》
 《国际海上危险货物运输》
 《Technical Instructions for the Safe Transport of Dangerous Goods》
 《危险货物安全运输技术指南》
 《Classification and code of dangerous goods》
 《危险货物分类与代码》
 《Occupational Safety and Health Act》(OSHA)
 《职业安全与健康法案》(OSHA)
 《Toxic Substance Control Act》(TSCA)
 《有毒物质控制法》(TSCA)
 《Consumer Product Safety Act》(CPSA)
 《消费者产品安全法案》(CPSA)
 《Federal Environmental Pollution Control Act》(FEPCA)
 《联邦环境污染控制法》(FEPCA)
 《The Oil Pollution Act》(OPA)
 《石油污染法》(OPA)
 《Superfund Amendments and Reauthorization Act TitleIII(302/311/312/313)》(SARA)
 《超级基金修正案和再授权法案 TitleIII(302/311/312/313)》(SARA)
 《Resource Conservation and Recovery Act》(RCRA)
 《资源保护和恢复法案》(RCRA)
 《Safety Drinking Water Act》(CWA)
 《安全饮用水法》(CWA)
 《California Proposition 65》
 《加州 65 号提案》
 《Code of Federal Regulations》(CFR)
 《联邦条例》(CFR)
 Regulation (EC) No. 1907/2006 on the Registration, Evaluation, Authorization and Restriction of Chemicals (REACH)
 关于化学品的注册、评估、授权和限制(EC)第 1907/2006 号规例
 In accordance with all Federal, State and local laws.
 符合所有联邦、州和地方法律。

16. Additional Information (其他信息)

This file is only effective to the batteries (BL1860B) provided by Guangdong Alfdeco Smart Energy Co., Ltd. which is manufactured by Guangdong Alfdeco Smart Energy Co., Ltd. The commissioner provides the composition information of batteries and promises its integrity and accuracy. Users should read this file carefully and use the batteries in correct method. Shenzhen Yilink Testing Co., LTD. doesn't assume responsibility for any damage or loss because of misuse of batteries.

本文件仅对由委托方广东艾弗得智慧能源有限公司提供的，并由广东艾弗得智慧能源有限公司生产的电池 (BL1860B)有效。该电池的成分信息由委托方提供并承诺其完整性和准确性。用户应仔细阅读此文件，并按照正确的方法使用电池，如因电池使用不当造成的损害或损失，深圳亿联检测有限公司不承担任何责任。

Important 注 意 事 项

1. Nobody is allowed to photocopy or partly photocopy this test report without written permission of YILINK.
未经本试验室书面同意，不得复制或部分地复制本报告。
2. The test report is invalid without the signatures of Ratifier, Reviewer and Testing engineer.
本报告书无批准人、审核人、及主检人签名无效。
3. The test report is invalid if altered.
本报告涂改无效。
4. Objections to the test report must be submitted to YILINK within 15 days.
对报告书若有异议，应于收到报告之日起 15 天内向本公司提出。
5. The test report is valid for the tested samples only.
本报告仅对测试样品有效。
6. The test data and results do not have social proof function.
检测数据和结果不具备社会证明性作用。

*****End of report 报告结束*****

锂电池/钠离子电池 UN38.3 试验概要 Test Summary

项目编号: PEKSZ202511119665

单位信息 Company Information					
委托单位 Consignor		广东艾弗得智慧能源有限公司 广东省东莞市清溪镇罗裙埔一路7号2号楼501室 电话/Tel: 0755-84571435 邮箱/ Mail: guobinsheng@jingnoo.com			
生产单位 Manufacturer		广东艾弗得智慧能源有限公司 广东省东莞市清溪镇罗裙埔一路7号2号楼501室 电话/Tel: 0755-84571435 邮箱/ Mail: guobinsheng@jingnoo.com			
测试单位 Test Lab		深圳亿联检测有限公司 深圳市坪山区龙田街道老坑社区锦绣中路19号美讯数码科技厂区1号厂房1406 电话/Tel: 86-755-89366579 邮箱/Mail: helen.chen@yilianmail.com 网址/Website: http://yi-link.cn			
电池信息 Battery Information					
名称 Name		可充电锂离子电池 RECHARGEABLE LITHIUM ION BATTERY		电池/电芯类别 Battery/Cell Classification	
型号 Type		BL1860B		锂离子電池 Li-ion Battery	
额定电压 Normal Voltage		18V		商标 Trademark	
额定能量 Watt-hour rating		108Wh		额定容量 Rated Capacity	
重量/Mass		570.7g		外观/Appearance	
				6Ah	
				青黑色近长方体 Cyan and Black almost cuboid	
				不适用 N/A	
测试信息 Test Information					
测试报告编号 Test Report Number		YL251145001U01		测试报告签发日期 Date of Test Report	
测试标准 Edition of UN Manual of Tests and Criteria Used		联合国《试验和标准手册》(第8版)38.3节 UN "Manual of Tests and Criteria" ST/SG/AC.10/11/Rev.8/Subsection 38.3			
T.1: 高度模拟 Altitude Simulation		通过 Pass	T.2: 温度试验 Thermal Test	通过 Pass	T.3: 振动 Vibration
T.4: 冲击 Shock		通过 Pass	T.5: 外部短路 External Short Circuit	通过 Pass	通过 Pass
T.7: 过度充电 Overcharge		通过 Pass	T.8: 强制放电 Forced Discharge	通过 Pass	T.6: 撞击/挤压 Impact/Crush
UN38.3.3.1(f)或/or UN38.3.3.2(d)		不适用 N/A		UN38.3.3.1(g)或/or UN38.3.3.2(e)	
UN38.3.3.1(f)或/or UN38.3.3.2(d)		不适用 N/A		不适用 N/A	
签名 Signatory 职务 Title		曾平平 检验员		签发日期 Issued Date	
				2025-11-11	



UN38.3 检测报告

UN38.3 Test Report

Client 委托方	Guangdong Alfdeco Smart Energy Co., Ltd 广东艾弗得智慧能源有限公司
Add. of Client 委托方地址	Room 501, Building 2, No.7, Luoqunpu 1st Road, Qingxi Town, Dongguan City, Guangdong Province 广东省东莞市清溪镇罗裙埔一路7号2号楼501室
Samples Description 样品名称	RECHARGEABLE LITHIUM ION BATTERY 可充电锂离子电池
Model/Type 型号规格	BL1860B, 18V 6Ah 108Wh
Testing Laboratory 测试机构:	Shenzhen Yilink Testing Co., LTD. 深圳亿联检测有限公司 Room 1406, Building A, Meixun Science Park, No.19 Jinxiu Middle Road, Laokeng Community, Longtian Street, Pingshan District, Shenzhen 深圳市坪山区龙田街道老坑社区锦绣中路19号美讯数码科技厂区1号厂房1406
Test Reference 测试依据	UNITED NATIONS "Manual of Tests and Criteria"(ST/SG/AC.10/11/Rev.8 Section 38.3) 联合国《试验和标准手册》第八版第38.3节
Report No. 报告编号	YL251145001U01
Issued Date 发行日期	2025.11.06
Test Conclusion 测试结论: Shown in the Conclusion of test report. 见检测报告结论页.	

Tested by 编写: Jack Tang

Approved by 批准: Jack Tang



Inspected by 审核人: Noah shaw

Date 日期: 2025.11.06

I、Sample Description 样品描述

Product Name 产品名称	RECHARGEABLE LITHIUM ION BATTERY 可充电锂离子电池		Sample Model 样品型号	BL1860B	
Manufacturer 制造商	Guangdong Alfdeco Smart Energy Co., Ltd 广东艾弗得智慧能源有限公司				
Address 地址	Room 501, Building 2, No.7, Luoqunpu 1st Road, Qingxi Town, Dongguan City, Guangdong Province 广东省东莞市清溪镇罗裙埔一路 7 号 2 号楼 501 室				
Factory 工厂	Guangdong Alfdeco Smart Energy Co., Ltd 广东艾弗得智慧能源有限公司				
Address 地址	Room 501, Building 2, No.7, Luoqunpu 1st Road, Qingxi Town, Dongguan City, Guangdong Province 广东省东莞市清溪镇罗裙埔一路 7 号 2 号楼 501 室				
Manufacturer' s contact information 制造商联系信息	Phone number 电话号码		Email address 电子邮箱地址		Website 网址
	0755-84571435		guobinsheng@jingnoo.com		--
Trade Mark 商标	--	Battery Shape 电池形状	Cuboid 长方体	Battery Size 电池尺寸 (L×W×T)	(117.0 x 75.0 x 67.0) mm
Nominal Voltage 标称电压	18V	Rated Capacity 额定容量	6Ah 108Wh	Limited Charge Voltage 充电限制电压	21V
Standard Charge Current 标准充电电流	3A	Maximum Continuous Charge Current 最大持续充电电流	6A	End Charge Current 结束充电电流	40mA
Cut-off Voltage 放电截止电压	13.5V	Standard Discharge Current 标准放电电流	1.2A	Maximum Discharge Current 最大放电电流	6A
Cell Number 组成电芯数量	10PCS		Cell Model 电芯型号	INR18650P	
Sample Mass 样品重量	570.7g		Sample Physical description 样品物理形态	Black and cyan almost cuboid 黑色和青色近长方体	
Receiving Date 接收日期	2025.10.24		Completing Date 完成日期	2025.11.06	

II、Standard 标准

UNITED NATIONS "Manual of Tests and Criteria" (ST/SG/AC.10/11/Rev.8 Section 38.3)

联合国《试验和标准手册》第八版第 38.3 节。

III、Test Item 测试项目

- | | |
|---|---|
| T.1. ☒ Altitude simulation 高度模拟 | T.5. ☒ External short circuit 外部短路 |
| T.2. ☒ Thermal test 温度试验 | T.6. ☒ Impact 撞击/ ☐ Crush 挤压 |
| T.3. ☒ Vibration 振动 | T.7. ☒ Overcharge 过充电 |
| T.4. ☒ Shock 冲击 | T.8. ☒ Forced discharge 强制放电 |

IV、Test Method and Requirement 测试方法和要求

Tests T.1 to T.5 shall be conducted in sequence on the same cell or battery. Tests T.6 and T.8 shall be conducted using not otherwise tested cells. Test T.7 may be conducted using undamaged batteries previously used in tests T.1 to T.5 for purposes of testing on cycled batteries.

用相同的电芯或电池按照顺序进行试验 T.1 至 T.5。试验 T.6 至 T.8 用没有进行其他试验的电芯。试验 T7 可以使用原先在试验 T1 至 T5 中使用过的未损坏的电池进行，以便测试交替充电放电的电池。

Batteries of B1#~B4# 、B9#~B12# are full charged after one cycle;

Batteries of B5#~B8# 、B13#~B16# are full charged after twenty-five cycles;

Component cells of C1#~C5# are 50% charged after one cycle;

Component cells of C6#~C10# are 50% charged after twenty-five cycles;

Component cells of C11#~C20# are full discharged after one cycle;

Component cells of C21#~C30# are full discharged after twenty-five cycles;

Test environment condition: ambient temperature: 15-25℃, ambient humidity: 40-70%

电池 B1#~B4# , B9#~B12#为 1 次循环满电状态;

电池 B5#~B8#, B13#~B16#为 25 次循环满电状态;

组成电芯 C1#~C5#为 1 次循环后 50%充电状态;

组成电芯 C6#~C10#为 25 次循环后 50%充电状态;

组成电芯 C11#~C20#为 1 次循环完全放电状态;

组成电芯 C21#~C30#为 25 次循环完全放电状态;

试验环境条件: 环境温度: 15-25℃, 环境湿度: 40-70%

In order to quantify the mass loss, the following procedure is provided:

$$\text{Mass loss (\%)} = (M1-M2)/M1 \times 100$$

质量损失的量化值, 可用以下公式计算:

$$\text{质量损失(\%)}=(M1-M2)/M1 \times 100$$

Where M1 is the mass before the test and M2 is the mass after the test. When mass loss does not exceed the values in Table below, it shall be considered as "no mass loss".

式中: M1 是试验前的质量, M2 是试验后的质量。如果质量损失不超过下表所列的数值, 应视为“无质量损失”。

Mass M of cell or battery 电芯或电池的质量	Mass loss limit 质量损失限值
$M < 1g$	0.5%
$1g \leq M \leq 75g$	0.2%
$M > 75g$	0.1%

Leakage means the visible escape of electrolyte or other material from a cell or battery or the loss of material (except battery casing, handling devices or labels) from a cell or battery such that the loss of mass exceeds the values in Table above.

渗漏系指可以看到的电解液或者其他物质从电芯或者电池中漏出，或电芯或电池中的物质损失（不包括电池外壳、搬运装置、或标签），失去的质量超过上表所列的数值。

In test T.1 to T.4, cells and batteries meet this requirement if there is no leakage, no venting, no disassembly, no rupture and no fire and if the open circuit voltage of each test cell or battery after testing is not less than 90% of its voltage immediately prior to this procedure. The requirement relating to voltage is not applicable to test cells and batteries at fully discharged states.

在测试 T.1 至 T.4 中，电芯和电池须满足无渗漏、无泄气、无解体、无破裂和无起火，并且每个试验电芯或电池在试验后的开路电压不小于其在进行这一试验前电压的 90%。

T.1. Altitude simulation 高度模拟

Test method 测试方法

Test cells and batteries are stored at a pressure of 11.6 kPa or less for at least six hours at ambient temperature ($20 \pm 5^\circ\text{C}$).

试验电芯和电池被放置在压力等于或低于 11.6 kPa 和环境温度($20 \pm 5^\circ\text{C}$)下存放至少 6 小时。

Requirement 要求

Cells and batteries meet this requirement if there is no leakage, no venting, no disassembly, no rupture and no fire and if the open circuit voltage of each test cell or battery after testing is not less than 90% of its voltage immediately prior to this procedure.

电芯和电池须无渗漏、无泄气、无解体、无破裂和无起火，并且每个试验电芯或电池在试验后的开路电压不小于其在进行这一试验前电压的 90%。

T.2. Thermal test 温度试验

Test method 测试方法

Test cells and batteries are to be stored for at least six hours at a test temperature equal to $72 \pm 2^\circ\text{C}$, followed by storage for at least six hours at a test temperature equal to $-40 \pm 2^\circ\text{C}$. The maximum time interval between test temperature extremes is 30 minutes. This procedure is to be repeated until 10 total cycles are complete, after which all test cells and batteries are to be stored for 24 hours at ambient temperature ($20 \pm 5^\circ\text{C}$). For large cells and batteries the duration of exposure to the test temperature extremes should be at least 12 hours.

试验电芯和电池放置在试验温度等于 $72 \pm 2^\circ\text{C}$ 的条件下存放至少 6 小时，接着再在试验温度等于 $-40 \pm 2^\circ\text{C}$ 的条件下存放至少 6 小时。两个极端试验温度之间的最大时间间隔为 30 分钟。此程序重复进行，共完成 10 次循环，接着将所有试验电芯和电池在环境温度($20 \pm 5^\circ\text{C}$)下存放 24 小时。对于大型电芯和电池，暴露于极端试验温度的时间至少应为 12 小时。

Requirement 要求

Cells and batteries meet this requirement if there is no leakage, no venting, no disassembly, no rupture

and no fire and if the open circuit voltage of each test cell or battery after testing is not less than 90% of its voltage immediately prior to this procedure.

电芯和电池须无渗漏、无泄气、无解体、无破裂和无起火，并且每个试验电芯或电池在试验后的开路电压不小于其在进行这一试验前电压的 90%。

T.3. Vibration 振动

Test method 测试方法

Cells and batteries are firmly secured to the platform of the vibration machine without distorting the cells in such a manner as to faithfully transmit the vibration. The vibration shall be a sinusoidal waveform with a logarithmic sweep between 7 Hz and 200 Hz and back to 7 Hz traversed in 15 minutes. This cycle shall be repeated 12 times for a total of 3 hours for each of three mutually perpendicular mounting positions of the cell. One of the directions of vibration must be perpendicular to the terminal face.

电芯和电池紧固于振动台台面，但不得造成电芯变形，并能准确可靠地传播振动。振动应是正弦波形，对数扫描频率在 7 Hz 和 200 Hz 之间，再回到 7 Hz，跨度为 15 分钟。这一振动过程须对三个互相垂直的电芯安装方位的每一方向重复进行 12 次，总共为时 3 小时。其中一个振动方向必须与端面垂直。

The logarithmic frequency sweep shall differ for cells and batteries with a gross mass of not more than 12 kg (cells and small batteries), and for batteries with a gross mass of more than 12 kg (large batteries).

作对数式频率扫描，对电芯和总质量不超过 12 千克的电池（电芯和小型电池），和对质量超过 12 千克的电池（大型电池）有所不同。

For cells and small batteries : from 7 Hz a peak acceleration of 1 gn is maintained until 18 Hz is reached. The amplitude is then maintained at 0.8 mm (1.6 mm total excursion) and the frequency increased until a peak acceleration of 8 gn occurs (approximately 50 Hz). A peak acceleration of 8 gn is then maintained until the frequency is increased to 200 Hz.

对电芯和小型电池：从 7 Hz 开始，保持 1 gn 的最大加速度，直到频率达到 18 Hz。然后将振幅保持在 0.8mm（总位移 1.6mm），并增加频率直到峰值加速度达到 8 gn（频率约为 50 Hz）。将峰值加速度保持在 8 gn 直到频率增加到 200 Hz。

For large batteries : from 7 Hz a peak acceleration of 1 gn is maintained until 18 Hz is reached. The amplitude is then maintained at 0.8 mm (1.6 mm total excursion) and the frequency increased until a peak acceleration of 2 gn occurs (approximately 25 Hz). A peak acceleration of 2 gn is then maintained until the frequency is increased to 200 Hz.

对大型电池：从 7 Hz 开始，保持 1 gn 的最大加速度，直到频率达到 18 Hz。然后将振幅保持在 0.8mm（总位移 1.6mm），并增加频率直到峰值加速度达到 2 gn（频率约为 25Hz）。将峰值加速度保持在 2 gn 直到频率增加到 200 Hz。

Requirement 要求

Cells and batteries meet this requirement if there is no leakage, no venting, no disassembly, no rupture and no fire and if the open circuit voltage of each test cell or battery after testing is not less than 90% of its voltage immediately prior to this procedure.

电芯和电池须无渗漏、无泄气、无解体、无破裂和无起火，并且每个试验电芯或电池在试验后的开路电压不小于其在进行这一试验前电压的 90%。

T.4. Shock 冲击

Test method 测试方法

Test cells and batteries shall be secured to the testing machine by means of a rigid mount which will support all mounting surfaces of each test battery.

试验电芯和电池用刚性支架紧固在试验装置上，支架支撑着每个试验电池的所有安装面。

Each cell shall be subjected to a half-sine shock of peak acceleration of 150 gn and pulse duration of 6 milliseconds. Alternatively, large cells may be subjects to a half-sine shock of peak acceleration of 50 gn and pulse duration of 11 milliseconds.

每个电芯须经受峰值加速度 150 gn 和脉冲持续时间 6 ms 的半正弦波冲击。不过，大型电芯须经受峰值

加速度 50 gn 和脉冲持续时间 11 ms 的半正弦波冲击。

Each battery shall be subjected to a half-sine shock of peak acceleration depending on the mass of the battery. The pulse duration shall be 6 milliseconds for small batteries and 11 milliseconds for large batteries. The formulas below are provided to calculate the appropriate minimum peak accelerations.

每个电池须经受半正弦波冲击,峰值加速度需要根据电池的重量来决定。小型电池的脉冲持续时间为 6 ms,大型电池的脉冲持续时间为 11ms。下面的公式是用来计算合适的最小峰值加速度。

电池	最小峰值加速度	脉冲持续时间
小型电池	150 gn 或计算结果中取最小的值 $\text{加速度 (gn)} = \sqrt{\left(\frac{100850}{\text{mass}}\right)}$	6ms
大型电池	50 gn 或计算结果中取最小的值 $\text{加速度 (gn)} = \sqrt{\left(\frac{30000}{\text{mass}}\right)}$	11 ms

Each cell or battery shall be subjected to three shocks in the positive direction and to three shocks in the negative direction in each of three mutually perpendicular mounting positions of the cell or battery for a total of 18 shocks.

每个电芯或电池须在三个互相垂直的电芯或电池安装方位的正方向经受三次冲击,接着在反方向经受三次冲击,总共经受 18 次冲击。

Requirement 要求

Cells and batteries meet this requirement if there is no leakage, no venting, no disassembly, no rupture and no fire and if the open circuit voltage of each test cell or battery after testing is not less than 90% of its voltage immediately prior to this procedure.

电芯和电池须无渗漏、无泄气、无解体、无破裂和无起火,并且每个试验电芯或电池在试验后的开路电压不小于其在进行这一试验前电压的 90%。

T.5. External short circuit 外部短路

Test method 测试方法

The cell or battery to be tested shall be heated for a period of time necessary to reach a homogeneous stabilized temperature of $57\pm4^{\circ}\text{C}$, measured on the external case. This period of time depends on the size and design of the cell or battery and should be assessed and documented. If this assessment is not feasible, the exposure time shall be at least 6 hours for small cells and small batteries, and 12 hours for large cells and large batteries. Then the cell or battery at $57\pm 4^{\circ}\text{C}$ shall be subjected to one short circuit condition with a total external resistance of less than 0.1 ohm.

试验电芯或电池需要加热一段时间,以使其外壳温度均匀稳定地达到 $57\pm4^{\circ}\text{C}$ 。加热时间的长短是由电芯或电池的尺寸和设计来决定的,这个加热时间需要评估并记录。如果这个加热时间不好评估的话,对于小电芯和小电池需要在此温度下放置至少 6 个小时,对于大电芯和大电池至少放置 12 个小时。然后使电芯或电池在 $57\pm4^{\circ}\text{C}$ 下经受总外电阻小于 0.1Ω 的短路条件。

This short circuit condition is continued for at least one hour after the cell or battery external case temperature has returned to $57\pm4^{\circ}\text{C}$, or in the case of the large batteries, has decreased by half of the maximum temperature increase observed during the test and remains below that value.

短路测试持续到电芯或电池外壳温度回到 $57\pm4^{\circ}\text{C}$ 后至少持续 1 小时,针对大电池,外壳温度需要下降到测试过程中监控到的最大温度的一半以下。

The short circuit and cooling down phases shall be conducted at least at ambient temperature.

短路测试和冷却阶段至少应该在环境温度下进行。

Requirement 要求

Cells and batteries meet this requirement if their external temperature does not exceed 170°C and there is no disassembly, no rupture and no fire during the test and within six hours after test.

电芯和电池外壳温度不超过 170℃，并且在试验过程中及试验后 6 小时内无解体、无破裂，无起火。

T.6. Impact / Crush 撞击/挤压

Test procedure – Impact (applicable to cylindrical cells not less than 18.0 mm in diameter)

测试步骤 – 撞击（适用于直径大于等于 18.0 毫米以上的圆柱形电芯）

The test sample cell or component cell is to be placed on a flat smooth surface. A 15.8 mm $\pm$ 0.1mm diameter, at least 6 cm long, or the longest dimension of the cell, whichever is greater, Type 316 stainless steel bar is to be placed across the centre of the sample. A 9.1 kg $\pm$ 0.1 kg mass is to be dropped from a height of 61 $\pm$ 2.5 cm at the intersection of the bar and sample in a controlled manner using a near frictionless, vertical sliding track or channel with minimal drag on the falling mass. The vertical track or channel used to guide the falling mass shall be oriented 90 degrees from the horizontal supporting surface.

试样电芯或电芯组件放在平坦光滑表面上，一根 316 型不锈钢棒横放在试样中心，钢棒直径 15.8 毫米 $\pm$ 0.1 毫米，长度至少 6 厘米，或电芯最长端的尺度，取二者之长者。将一块 9.1 千克 $\pm$ 0.1 千克的重锤从 61 $\pm$ 2.5 厘米高度跌落到钢棒和试样交叉处，使用一个几乎没有摩擦的、对落体重锤阻力最小的垂直轨道或管道加以控制。垂直轨道或管道用于引导落锤沿与水平支撑表面呈 90 度落下。

The test sample is to be impacted with its longitudinal axis parallel to the flat surface and perpendicular to the longitudinal axis of the 15.8 mm $\pm$ 0.1mm diameter curved surface lying across the centre of the test sample. Each sample is to be subjected to only a single impact.

受撞击的试样，纵轴应与平坦表面平行并与横放在试样中心的直径 15.8 $\pm$ 0.1 毫米弯曲表面的纵轴垂直。每一试样只经受一次撞击。

Test procedure – Crush (applicable to prismatic, pouch, coin/button cells and cylindrical cells less than 18.0 mm in diameter)

测试步骤– 挤压（适用于棱柱形，袋状，硬币/纽扣电芯和圆柱形电芯直径小于 18.0 毫米）

A cell or component cell is to be crushed between two flat surfaces. The crushing is to be gradual with a speed of approximately 1.5 cm/s at the first point of contact. The crushing is to be continued until the first of the three options below is reached.

将电芯或电芯组件放在两个平面之间挤压，挤压力度逐渐加大，在第一个接触点上的速度大约为 1.5 cm/s。挤压持续进行，直到出现以下三种情况之一：

- (a) The applied force reaches 13 kN $\pm$ 0.78 kN;
 - (b) The voltage of the cell drops by at least 100 mV;
 - (c) The cell is deformed by 50% or more of its original thickness.
- (a)施加的力达到 13 kN $\pm$ 0.78 kN;
- (b)电芯的电压下降至少 100mV;
- (c)电芯形变达到原始厚度的 50%或更多。

Once the maximum pressure has been obtained, the voltage drops by 100 mV or more, or the cell is deformed by at least 50% of its original thickness, the pressure shall be released.

一旦达到最大压力、电压下降 100mV 或更多，或电芯形变至少达到原始厚度的 50%，即可解除压力。

A prismatic or pouch cell shall be crushed by applying the force to the widest side. A button/coin cell shall be crushed by applying the force on its flat surfaces. For cylindrical cells, the crush force shall be applied perpendicular to the longitudinal axis.

棱柱形或袋装电芯须从最宽的面施压。纽扣/硬币形电芯应从其平坦表面施压。圆柱形电芯应从与纵轴垂直的方向施压。

Each test cell or component cell is to be subjected to one crush only. The test sample shall be observed for a further 6 h. The test shall be conducted using test cells or component cells that have not previously been subjected to other tests.

每个试样电芯或电芯组件只做一次挤压试验。试样须继续观察 6 小时。试验须使用之前未做过其他试验的试样电芯或电芯组件进行。

Requirement 要求

Cell and component cells meet this requirement if their external temperature does not exceed 170°C and there is no disassembly and no fire during the test and within six hours after test.

电芯和电芯组件外壳温度不超过 170°C，并且在试验过程中及试验后 6 小时内无解体，无起火。

T.7. Overcharge 过充电

Test method 测试方法

The charge current shall be twice the manufacturer's recommended maximum continuous charge current. The minimum voltage of the test shall be as follows:

充电电流为制造商推荐的最大持续充电电流的两倍。试验的最小电压如下：

- (a) When the manufacturer's recommended charge voltage is not more than 18V, the minimum voltage of the test shall be the lesser of two times the maximum charge voltage of the battery or 22V.
- (b) When the manufacturer's recommended charge voltage is more than 18V, the minimum voltage of the test shall be 1.2 times the maximum charge voltage.
- (a) 制造商推荐的充电电压不大于 18 伏时，试验的最小电压应是电池最大充电电压的两倍或 22 伏两者中的较小者。
- (b) 制造商推荐的充电电压大于 18 伏时，试验的最小电压应是电池最大充电电压的 1.2 倍。

Tests are to be conducted at ambient temperature. The duration of the test shall be 24 hours.

试验应在环境温度下进行。进行试验的时间应为 24 小时。

Requirement 要求

Rechargeable batteries meet this requirement if there is no disassembly and no fire during the test and within seven days after the test.

充电电池应在试验过程中和试验后 7 天内无解体，无起火。

T.8. Forced discharge 强制放电

Test method 测试方法

Each cell shall be forced discharged at ambient temperature by connecting it in series with a 12V D.C. power supply at an initial current equal to the maximum discharge current specified by the manufacturer.

每个电芯在环境温度下与 12V 直流电电源串联在起始电流等于制造商给定的最大放电电流的条件下强制放电。

The specified discharge current is to be obtained by connecting a resistive load of the appropriate size and rating in series with the test cell. Each cell is forced discharged for a time interval (in hours) equal to its rated capacity divided by the initial test current (in ampere).

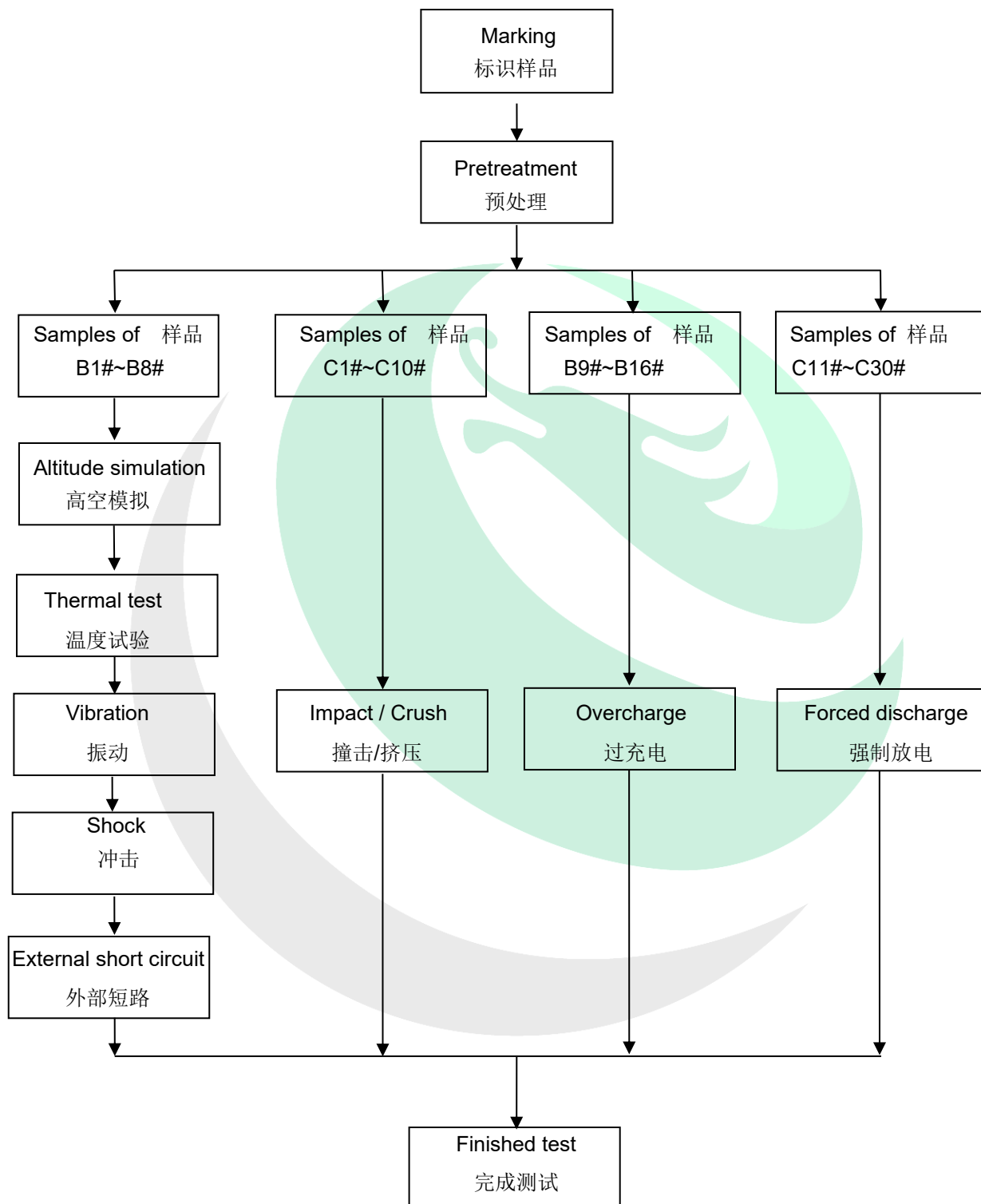
试样电芯与一个适当大小的电阻负载串联以调节到规定大小的放电电流。每个电芯的放电时间（单位为 h）等于电芯的额定容量除以试验初始放电电流（单位 A）。

Requirement 要求

Primary or rechargeable cells meet this requirement if there is no disassembly and no fire during the test and within seven days after the test.

原电芯或充电电芯应在试验过程中和试验后 7 天内无解体，无起火。

V、Test Procedure 测试流程



VI、Test Data 测试数据

T.1. Altitude simulation 高度模拟

The state of cells 样品状态	No. 编号	Pre-test 试验前		After test 试验后		Mass loss 质量损失 (%)	Voltage after test/Voltage pre-test 试验后电压/试验前电压(%)	Status 结果
		Mass 质量 (g)	Voltage 电压 (V)	Mass 质量 (g)	Voltage 电压 (V)			
Full charged after one cycle 1次循环后满电状态	B1#	570.62	20.87	570.60	20.86	0.004	99.952	Pass 合格
	B2#	570.66	20.90	570.64	20.88	0.004	99.904	Pass 合格
	B3#	570.60	20.86	570.57	20.85	0.005	99.952	Pass 合格
	B4#	570.63	20.88	570.61	20.86	0.004	99.904	Pass 合格
Full charged after twenty-five cycles 25次循环后满电状态	B5#	570.61	20.87	570.60	20.85	0.002	99.904	Pass 合格
	B6#	570.68	20.91	570.66	20.90	0.004	99.952	Pass 合格
	B7#	570.64	20.89	570.62	20.87	0.004	99.904	Pass 合格
	B8#	570.64	20.90	570.61	20.88	0.005	99.904	Pass 合格
Notes 注释: Atmospheric pressure 大气压强: $1.013 \times 10^5 \text{Pa}$, Ambient temperature 环境温度: 23.3°C After the test, there is no leakage, no venting, no disassembly, no rupture and no fire. 测试后, 电池未渗漏、未泄气、未解体、未破裂和未起火。								

T.2. Thermal test 温度试验

The state of cells 样品状态	No. 编号	Pre-test 试验前		After test 试验后		Mass loss 质量损失 (%)	Voltage after test/Voltage pre-test 试验后电压/试验前电压(%)	Status 结果
		Mass 质量 (g)	Voltage 电压 (V)	Mass 质量 (g)	Voltage 电压 (V)			
Full charged after one cycle 1次循环后满电状态	B1#	570.60	20.86	570.51	20.66	0.016	99.041	Pass 合格
	B2#	570.64	20.88	570.54	20.67	0.018	98.994	Pass 合格
	B3#	570.57	20.85	570.48	20.65	0.016	99.041	Pass 合格
	B4#	570.61	20.86	570.53	20.67	0.014	99.089	Pass 合格
Full charged after twenty-five cycles 25次循环后满电状态	B5#	570.60	20.85	570.51	20.64	0.016	98.993	Pass 合格
	B6#	570.66	20.90	570.56	20.71	0.018	99.091	Pass 合格
	B7#	570.62	20.87	570.53	20.67	0.016	99.042	Pass 合格
	B8#	570.61	20.88	570.53	20.69	0.014	99.090	Pass 合格
Notes 注释: Atmospheric pressure 大气压强: $1.013 \times 10^5 \text{Pa}$, Ambient temperature 环境温度: 23.2°C After the test, there is no leakage, no venting, no disassembly, no rupture and no fire. 测试后, 电池未渗漏、未泄气、未解体、未破裂和未起火。								

T.3. Vibration 振动

The state of cells 样品状态	No. 编号	Pre-test 试验前		After test 试验后		Mass loss 质量损失 (%)	Voltage after test/Voltage pre-test 试验后电压/试验前电压(%)	Status 结果
		Mass 质量 (g)	Voltage 电压 (V)	Mass 质量 (g)	Voltage 电压 (V)			
Full charged after one cycle 1次循环后满电状态	B1#	570.51	20.66	570.49	20.63	0.004	99.855	Pass 合格
	B2#	570.54	20.67	570.52	20.64	0.004	99.855	Pass 合格
	B3#	570.48	20.65	570.47	20.61	0.002	99.806	Pass 合格
	B4#	570.53	20.67	570.52	20.64	0.002	99.855	Pass 合格
Full charged after twenty-five cycles 25次循环后满电状态	B5#	570.51	20.64	570.50	20.61	0.002	99.855	Pass 合格
	B6#	570.56	20.71	570.54	20.68	0.004	99.855	Pass 合格
	B7#	570.53	20.67	570.51	20.65	0.004	99.903	Pass 合格
	B8#	570.53	20.69	570.52	20.66	0.002	99.855	Pass 合格
Notes 注释: Atmospheric pressure 大气压强: $1.013 \times 10^5 \text{Pa}$, Ambient temperature 环境温度: 23.7°C After the test, there is no leakage, no venting, no disassembly, no rupture and no fire. 测试后, 电池未渗漏、未泄气、未解体、未破裂和未起火。								

T.4. Shock 冲击

The state of cells 样品状态	No. 编号	Pre-test 试验前		After test 试验后		Mass loss 质量损失 (%)	Voltage after test/Voltage pre-test 试验后电压/试验前电压(%)	Status 结果
		Mass 质量 (g)	Voltage 电压 (V)	Mass 质量 (g)	Voltage 电压 (V)			
Full charged after one cycle 1次循环后满电状态	B1#	570.49	20.63	570.47	20.62	0.004	99.952	Pass 合格
	B2#	570.52	20.64	570.50	20.62	0.004	99.903	Pass 合格
	B3#	570.47	20.61	570.46	20.60	0.002	99.951	Pass 合格
	B4#	570.52	20.64	570.51	20.62	0.002	99.903	Pass 合格
Full charged after twenty-five cycles 25次循环后满电状态	B5#	570.50	20.61	570.49	20.60	0.002	99.951	Pass 合格
	B6#	570.54	20.68	570.52	20.66	0.004	99.903	Pass 合格
	B7#	570.51	20.65	570.50	20.63	0.002	99.903	Pass 合格
	B8#	570.52	20.66	570.51	20.65	0.002	99.952	Pass 合格
Notes 注释: Atmospheric pressure 大气压强: $1.013 \times 10^5 \text{Pa}$, Ambient temperature 环境温度: 23.5°C After the test, there is no leakage, no venting, no disassembly, no rupture and no fire. 测试后, 电池未渗漏、未泄气、未解体、未破裂和未起火。								

T.5. External short circuit 外部短路

The state of cells 样品状态	No. 编号	External Peak temperature(°C) 电池表面最高温度(°C)	Status 结果
Full charged after one cycle 1 次循环后满电状态	B1#	57.8	Pass 合格
	B2#	58.3	Pass 合格
	B3#	58.6	Pass 合格
	B4#	58.2	Pass 合格
Full charged after twenty-five cycles 25 次循环后满电状态	B5#	57.9	Pass 合格
	B6#	57.8	Pass 合格
	B7#	59.2	Pass 合格
	B8#	59.1	Pass 合格
Notes 注释: Atmospheric pressure 大气压强: $1.013 \times 10^5 \text{Pa}$, Ambient temperature 环境温度: 23.3°C There is no disassembly, no rupture and no fire during the test and within six hours after test. 电池在测试中和测试后 6 小时内未解体、未破裂, 未起火。			

T.6. Impact 撞击

The state of cells 样品状态	No. 编号	External Peak temperature(°C) 电池表面最高温度(°C)	Status 结果
50% charged after one cycle 1 次循环后 50%充电状态	C1#	108.1	Pass 合格
	C2#	111.9	Pass 合格
	C3#	109.6	Pass 合格
	C4#	111.1	Pass 合格
	C5#	108.9	Pass 合格
50% charged after twenty-five cycles 25 次循环后 50%充电状态	C6#	112.0	Pass 合格
	C7#	111.2	Pass 合格
	C8#	109.1	Pass 合格
	C9#	111.8	Pass 合格
	C10#	109.9	Pass 合格
Notes 注释: Atmospheric pressure 大气压强: $1.013 \times 10^5 \text{Pa}$, Ambient temperature 环境温度: 23.2°C There is no disassembly and no fire during the test and within six hours after test. 电芯在测试中和测试后 6 小时内未解体、未起火。			

T.7. Overcharge 过充电

The state of cells 样品状态	No. 编号	Status 结果
Full charged after one cycle 1 次循环后满电状态	B9#	Pass 合格
	B10#	Pass 合格
	B11#	Pass 合格
	B12#	Pass 合格
Full charged after twenty-five cycles 25 次循环后满电状态	B13#	Pass 合格
	B14#	Pass 合格
	B15#	Pass 合格
	B16#	Pass 合格
Notes 注释: Atmospheric pressure 大气压强: $1.013 \times 10^5 \text{Pa}$, Ambient temperature 环境温度: 23.1°C There is no disassembly and no fire during the test and within seven days after the test. 电池在测试中和测试后 7 天内未解体, 未起火。		

T.8. Forced discharge 强制放电

The state of cells 样品状态	No. 编号	Status 结果
Full discharged after one cycle 1 次循环完全放电状态	C11#	Pass 合格
	C12#	Pass 合格
	C13#	Pass 合格
	C14#	Pass 合格
	C15#	Pass 合格
	C16#	Pass 合格
	C17#	Pass 合格
	C18#	Pass 合格
	C19#	Pass 合格
	C20#	Pass 合格
Full discharged after twenty-five cycles 25 次循环完全放电状态	C21#	Pass 合格
	C22#	Pass 合格
	C23#	Pass 合格
	C24#	Pass 合格
	C25#	Pass 合格
	C26#	Pass 合格

	C27#	Pass 合格
	C28#	Pass 合格
	C29#	Pass 合格
	C30#	Pass 合格
Notes 注释: Atmospheric pressure 大气压强: $1.013 \times 10^5 \text{Pa}$, Ambient temperature 环境温度: 23.4°C There is no disassembly and no fire during the test and within seven days after the test. 电芯在测试中和测试后 7 天内未解体, 未起火。		



VII、Conclusion 结论

No. 编号	Test item 测试项目	Sample number 样品数量	Test reference 测试参考	Conclusion 结论
1	Altitude simulation 高空模拟	B1#~B8#	UN Manual of Test and Criteria, part III, subsection 38.3.4.1 UN 试验和标准手册,第III部分,第 38.3.4.1 节	Pass 合格
2	Thermal test 温度试验		UN Manual of Test and Criteria, part III, subsection 38.3.4.2 UN 试验和标准手册,第III部分,第 38.3.4.2 节	Pass 合格
3	Vibration 振动		UN Manual of Test and Criteria, part III, subsection 38.3.4.3 UN 试验和标准手册,第III部分,第 38.3.4.3 节	Pass 合格
4	Shock 冲击		UN Manual of Test and Criteria, part III, subsection 38.3.4.4 UN 试验和标准手册,第III部分,第 38.3.4.4 节	Pass 合格
5	External short circuit 外部短路		UN Manual of Test and Criteria, part III, subsection 38.3.4.5 UN 试验和标准手册,第III部分,第 38.3.4.5 节	Pass 合格
6	Impact/Crush 撞击/挤压	C1#~C10#	UN Manual of Test and Criteria, part III, subsection 38.3.4.6 UN 试验和标准手册,第III部分,第 38.3.4.6 节	Pass 合格
7	Overcharge 过度充电	B9#~B16#	UN Manual of Test and Criteria, part III, subsection 38.3.4.7 UN 试验和标准手册,第III部分,第 38.3.4.7 节	Pass 合格
8	Forced discharge 强制放电	C11#~C30#	UN Manual of Test and Criteria, part III, subsection 38.3.4.8 UN 试验和标准手册,第III部分,第 38.3.4.8 节	Pass 合格

The submitted samples were complied with the stated requirements of UN manual of test and criteria, part III, subsection 38.3, the test result is qualified.

经检测，提交的测试样品均符合 UN38.3 的要求，测试结论为合格。

VIII、Photo of The Sample 样品图片

Model 型号: BL1860B



Photo 1 Front 正面



Photo 2 Rear 反面

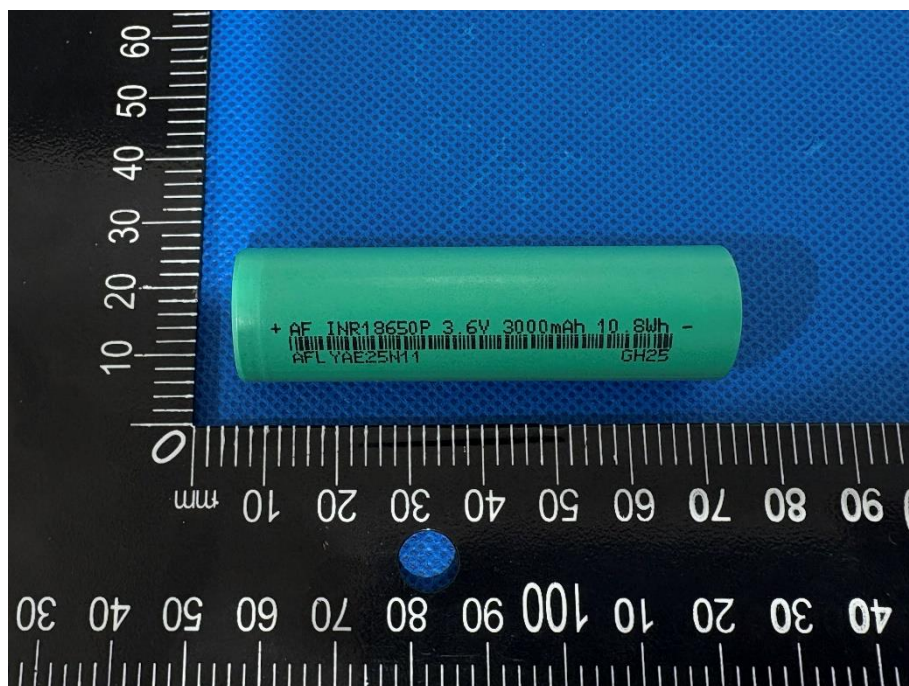


Photo 3 Internal Cell 内部电芯

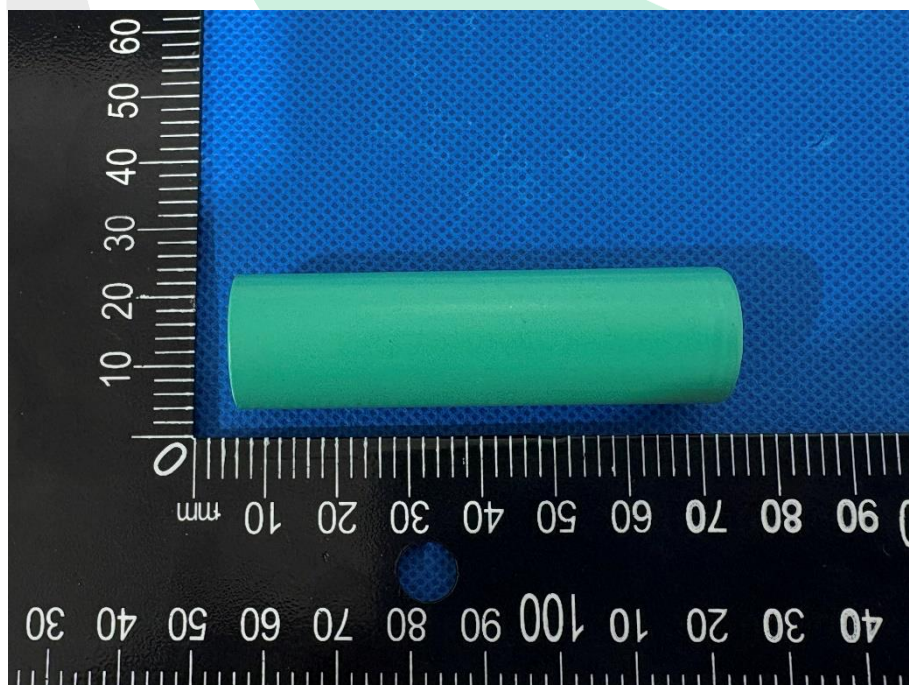


Photo 4 Internal Cell 内部电芯

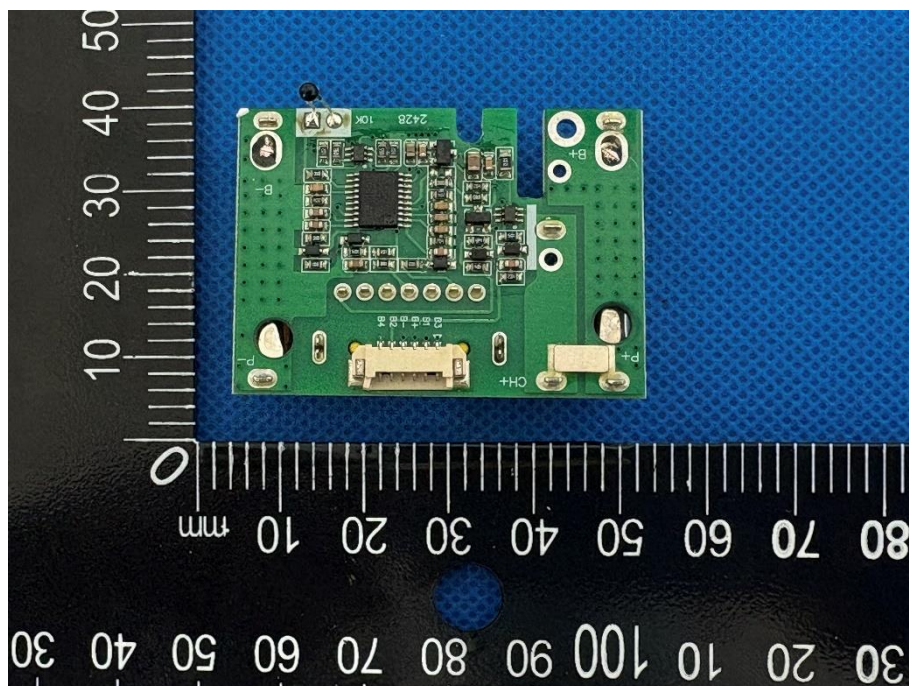


Photo 5 Protection board 保护板

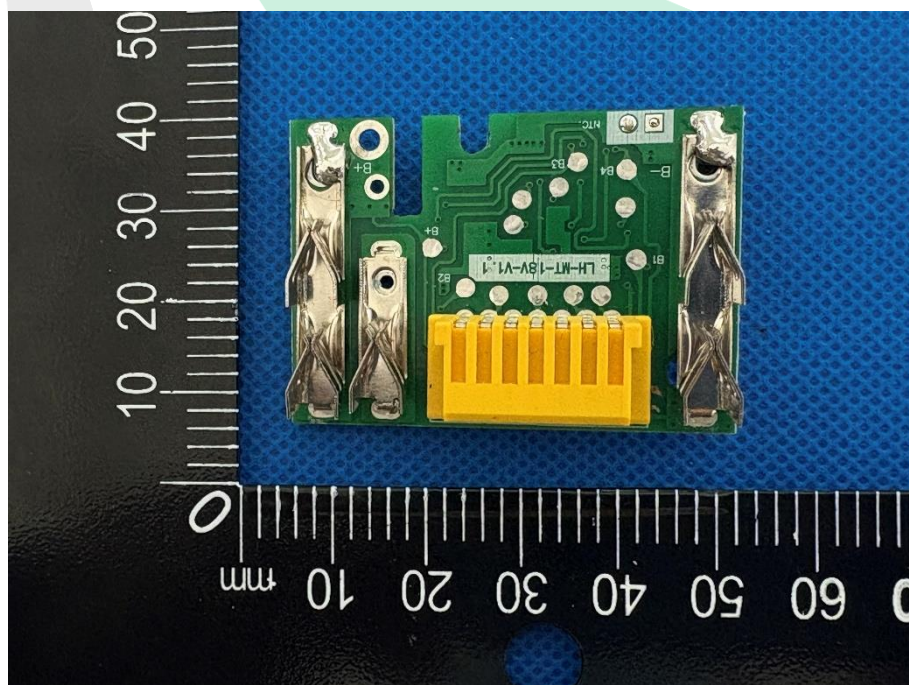


Photo 6 Protection board 保护板

Important 注 意 事 项

1. The test report is invalid without the official stamp of YILINK.
报告无检测单位印章无效。
2. Nobody is allowed to photocopy or partly photocopy this test report without written permission of YILINK.
未经本试验室书面同意，不得复制或部分地复制本报告。
3. The test report is invalid without the signatures of Ratifier, Reviewer and Testing engineer.
本报告书无批准人、审核人、及主检人签名无效。
4. The test report is invalid if altered.
本报告涂改无效。
5. Objections to the test report must be submitted to YILINK within 15 days.
对报告书若有异议，应于收到报告之日起 15 天内向本公司提出。
6. The test report is valid for the tested samples only.
本报告仅对测试样品有效。

*****End of report 报告结束*****

TEST REPORT IEC 62133-2

**Secondary cells and batteries containing alkaline or other non-acid electrolytes –
Safety requirements for portable sealed secondary cells, and for batteries made
from them, for use in portable applications-**

Part 2: Lithium systems

Report Number.: YL251145002

Tested by (+ signature): Cathy Lv

Reviewed by (+ signature): Noah Shao

Approved by (+ signature): Noah Shao

Date of issue: 2025-11-06

Total number of pages: 25 pages

Applicant's name: Guangdong Alfdeco Smart Energy Co., Ltd

Address: Room 501, Building 2, No.7, Luoqunpu 1st Road, Qingxi Town,
Dongguan City, Guangdong Province

Test specification:

Standard: IEC 62133-2:2017, IEC 62133-2:2017/AMD1:2021

Test procedure: Test Report

Non-standard test method: N/A

Test item description: RECHARGEABLE LITHIUM ION BATTERY

Trade Mark: N/A

Manufacturer: Guangdong Alfdeco Smart Energy Co., Ltd

Address: Room 501, Building 2, No.7, Luoqunpu 1st Road, Qingxi Town,
Dongguan City, Guangdong Province

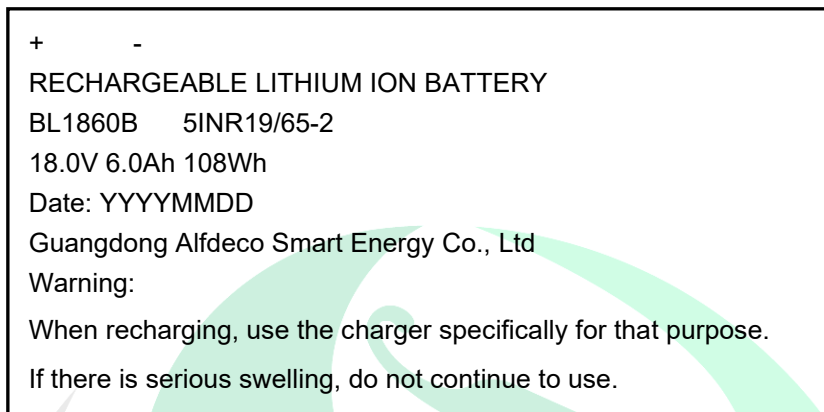
Model/Type reference: BL1860B

Ratings: 18.0V, 6.0Ah, 108Wh

Testing procedure and testing location:	
Testing Laboratory:	
Testing location/ address.....:	Shenzhen Yilink Testing Co., LTD. Room 1406, Building A, Meixun Science Park, No.19 Jinxiu Middle Road, Laokeng Community, Longtian Street, Pingshan District, Shenzhen
List of Attachments:	
Appendix 1: 3 pages of Photo Documentation	
Summary of testing:	
Tests performed (name of test and test clause): cl.5.6.2 Design recommendation; cl.7.1 Charging procedure for test purposes (for Cells and Batteries); cl.7.2.1 Continuous charging at constant voltage (cells); cl.7.2.2 Case stress at high ambient temperature (battery); cl.7.3.1 External short circuit (cells); cl.7.3.2 External short circuit (batteries); cl.7.3.3 Free fall (cells and batteries); cl.7.3.4 Thermal abuse (cells); cl.7.3.5 Crush (cells); cl.7.3.6 Over-charging of battery; cl.7.3.7 Forced discharge (cells); cl.7.3.8 Mechanical tests (batteries); cl.7.3.9 Design evaluation – Forced internal short circuit (cells). Tests are made with the number of cells and batteries specified in IEC 62133-2:2017 Table 1.	Testing location: Shenzhen Yilink Testing Co., LTD. Room 1406, Building A, Meixun Science Park, No.19 Jinxiu Middle Road, Laokeng Community, Longtian Street, Pingshan District, Shenzhen
Summary of compliance with National Differences	
N/A	
☒ The product fulfils the requirements of <u>EN 62133-2: 2017/A1:2021</u>	

Copy of marking plate

The artwork below may be only a draft. The use of certification marks on a product must be authorized by the respective NCBs that own these marks.



Remark:

For the date code "YYYYMMDD":

"YYYY" means the year of manufacture

"MM" means the month of manufacture

"DD" means the day of manufacture

Test item particulars	
Classification of installation and use	To be defined in final product
Supply connection	Special connector
Recommend charging method declared by the manufacturer	Charging the battery with 3000mA constant current until 21.0V, then constant voltage until charge current reduces to 40mA at ambient 20°C±5°C.
Discharge current (0,2 I_L A)	1200mA
Specified final voltage	13.5V
Upper limit charging voltage per cell	4.25V
Maximum charging current	3000mA
Charging temperature upper limit	45°C
Charging temperature lower limit	0°C
Polymer cell electrolyte type	☐ gel polymer ☐ solid polymer ☒ N/A
Possible test case verdicts:	
- test case does not apply to the test object	N/A
- test object does meet the requirement	P (Pass)
- test object does not meet the requirement	F (Fail)
Testing	
Date of receipt of test item	2025-10-24
Date (s) of performance of tests	2025-10-24 to 2025-11-06
General remarks:	
The test results presented in this report relate only to the object tested.	
This report shall not be reproduced, except in full, without the written approval of the Issuing testing laboratory.	
"(See Enclosure #)" refers to additional information appended to the report.	
"(See appended table)" refers to a table appended to the report.	
Throughout this report a ☐ comma / ☒ point is used as the decimal separator.	
Name and address of factory (ies) Same as manufacturer	

General product information:

This battery is constructed with ten lithium-ion cells (5S2P) and has overcharge, over-discharge, over current and short-circuits proof circuit.

The main features of the battery pack are shown as below (clause 7.1.1):

Model	Nominal capacity	Nominal voltage	Nominal Charge Current	Nominal Discharge Current	Maximum Charge Current	Maximum Discharge Current	Maximum Charge Voltage	Cut-off Voltage
BL1860B	6000mAh	18.0V	3000mA	1200mA	3000mA	6000mA	21.0V	13.5V

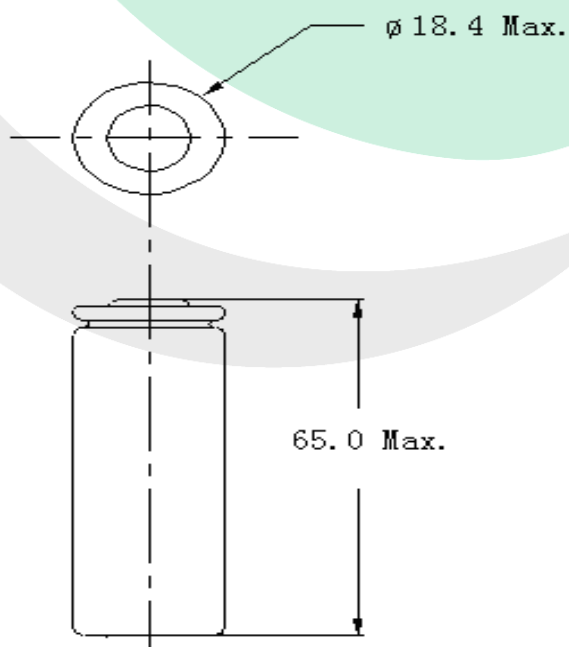
The main features of the cell in the battery pack are shown as below (clause 7.1.1):

Model	Nominal capacity	Nominal voltage	Nominal Charge Current	Nominal Discharge Current	Maximum Charge Current	Maximum Discharge Current	Maximum Charge Voltage	Cut-off Voltage
INR18650P	3000mAh	3.6V	600mA	600mA	1500mA	3000mA	4.25V	2.7V

The main features of the cell in the battery pack are shown as below (clause 7.1.2):

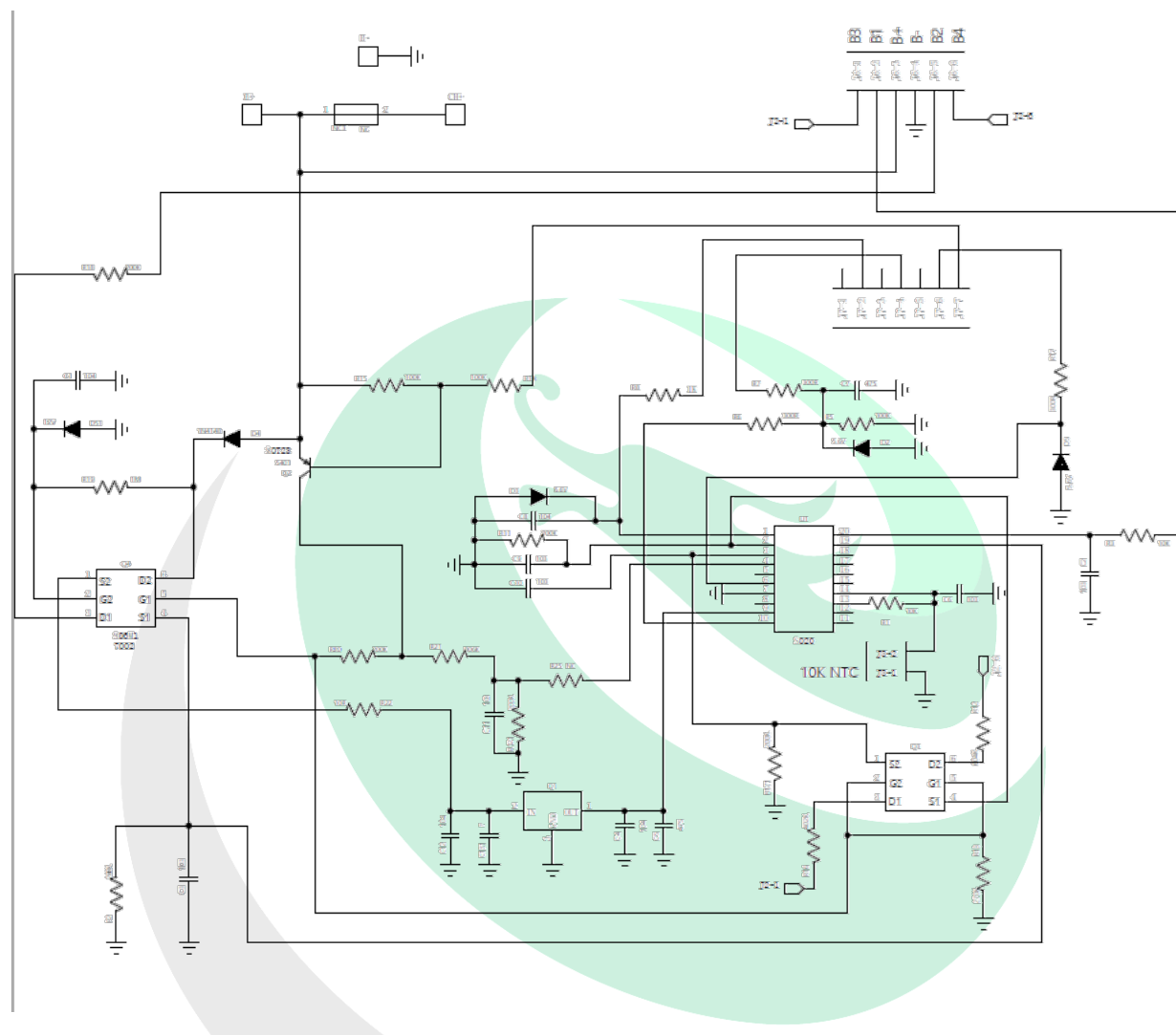
Model	Upper limit charge voltage	Taper-off current	Lower charge temperature	Upper charge temperature
INR18650P	4.25V	150mA	0°C	45°C

Construction:



Cell (Unit: mm)

Circuit diagram:



IEC 62133-2			
Clause	Requirement + Test	Result - Remark	Verdict
4	PARAMETER MEASUREMENT TOLERANCES		P
	Parameter measurement tolerances		P
5	GENERAL SAFETY CONSIDERATIONS		P
5.1	General		P
	Cells and batteries so designed and constructed that they are safe under conditions of both intended use and reasonably foreseeable misuse		P
5.2	Insulation and wiring		P
	The insulation resistance between the positive terminal and externally exposed metal surfaces of the battery (excluding electrical contact surfaces) is not less than 5 MΩ	No metal surface exists.	N/A
	Insulation resistance (MΩ)		—
	Internal wiring and insulation are sufficient to withstand maximum anticipated current, voltage and temperature requirements		P
	Orientation of wiring maintains adequate clearances and creepage distances between conductors		P
	Mechanical integrity of internal connections accommodates reasonably foreseeable misuse		P
5.3	Venting		P
	Battery cases and cells incorporate a pressure relief mechanism or are constructed so that they relieve excessive internal pressure at a value and rate that will preclude rupture, explosion and self-ignition		P
	Encapsulation used to support cells within an outer casing does not cause the battery to overheat during normal operation nor inhibit pressure relief		P
5.4	Temperature, voltage and current management		P
	Batteries are designed such that abnormal temperature rise conditions are prevented	Overcharge, overdischarge, over current and short-circuit proof circuit used in this battery. See tests of clause 7.	P
	Batteries are designed to be within temperature, voltage and current limits specified by the cell manufacturer	See above.	P
	Batteries are provided with specifications and charging instructions for equipment manufacturers so that specified chargers are designed to maintain charging within the temperature, voltage and current limits specified	The charging limits specified in the manufacturer's specification.	P
5.5	Terminal contacts		P
	The size and shape of the terminal contacts ensure that they can carry the maximum anticipated current	Special connector used.	P

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Clause	Requirement + Test	Result - Remark	Verdict
	External terminal contact surfaces are formed from conductive materials with good mechanical strength and corrosion resistance		P
	Terminal contacts are arranged to minimize the risk of short circuits		P
5.6	Assembly of cells into batteries		P
5.6.1	General		P
	Each battery has an independent control and protection for current, voltage, temperature and any other parameter required for safety and to maintain the cells within their operating region	5S2P, Protective circuit equipped on battery.	P
	This protection may be provided external to the battery such as within the charger or the end devices		N/A
	If protection is external to the battery, the manufacturer of the battery provide this safety relevant information to the external device manufacturer for implementation		N/A
	If there is more than one battery housed in a single battery case, each battery has protective circuitry that can maintain the cells within their operating regions		N/A
	Manufacturers of cells specify current, voltage and temperature limits so that the battery manufacturer/designer may ensure proper design and assembly	Current, voltage and temperature limits specified by cell manufacturer.	P
	Batteries that are designed for the selective discharge of a portion of their series connected cells incorporate circuitry to prevent operation of cells outside the limits specified by the cell manufacturer		N/A
	Protective circuit components are added as appropriate and consideration given to the end-device application		P
	The manufacturer of the battery provide a safety analysis of the battery safety circuitry with a test report including a fault analysis of the protection circuit under both charging and discharging conditions confirming the compliance	Safety analysis report provided by manufacturer.	P
5.6.2	Design recommendation		P
	For the battery consisting of a single cell or a single cellblock, it is recommended that the charging voltage of the cell does not exceed the upper limit of the charging voltage specified in Table 2		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	For the battery consisting of series-connected plural single cells or series-connected plural cellblocks, it is recommended that the voltages of any one of the single cells or single cellblocks does not exceed the upper limit of the charging voltage, specified in Table 2, by monitoring the voltage of every single cell or the single cellblocks	5S2P, Max. Charging voltage of each cell: 4.25V, not exceed 4.25V specified in Clause 7.1.2, Table 2.	P
	For the battery consisting of series-connected plural single cells or series-connected plural cellblocks, it is recommended that charging is stopped when the upper limit of the charging voltage is exceeded for any one of the single cells or single cellblocks by measuring the voltage of every single cell or the single cellblocks		N/A
	For batteries consisting of series-connected cells or cell blocks, nominal charge voltage are not counted as an overcharge protection		P
	For batteries consisting of series-connected cells or cell blocks, cells have closely matched capacities, be of the same design, be of the same chemistry and be from the same manufacturer		P
	It is recommended that the cells and cell blocks are not discharged beyond the cell manufacturer's specified final voltage		P
	For batteries consisting of series-connected cells or cell blocks, cell balancing circuitry are incorporated into the battery management system	Final voltage of battery: 13.5V, 2.7V for each cell, not exceed the final voltage specified by cell manufacturer.	P
5.6.3	Mechanical protection for cells and components of batteries		P
	Mechanical protection for cells, cell connections and control circuits within the battery are provided to prevent damage as a result of intended use and reasonably foreseeable misuse	Mechanical protection for cell connections and control circuits provided.	P
	The mechanical protection can be provided by the battery case or it can be provided by the end product enclosure for those batteries intended for building into an end product		P
	The battery case and compartments housing cells are designed to accommodate cell dimensional tolerances during charging and discharging as recommended by the cell manufacturer		P
	For batteries intended for building into a portable end product, testing with the battery installed within the end product is considered when conducting mechanical tests		N/A
5.7	Quality plan		P

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Clause	Requirement + Test	Result - Remark	Verdict
	The manufacturer prepares and implements a quality plan that defines procedures for the inspection of materials, components, cells and batteries and which covers the whole process of producing each type of cell or battery	Complied. Quality plan provided.	P
5.8	Battery safety components	See TABLE: Critical components information	N/A

6	TYPE TEST AND SAMPLE SIZE		P
	Tests are made with the number of cells or batteries specified in Table 1 using cells or batteries that are not more than six months old		P
	The internal resistance of coin cells are measured in accordance with Annex D. Coin cells with internal resistance less than or equal to 3 Ω are tested in accordance with Table 1	Not coin cells	N/A
	Unless otherwise specified, tests are carried out in an ambient temperature of 20 °C $\pm$ 5 °C		P
	The safety analysis of 5.6.1 identify those components of the protection circuit that are critical for short-circuit, overcharge and over discharge protection		P
	When conducting the short-circuit test, consideration is given to the simulation of any single fault condition that is likely to occur in the protecting circuit that would affect the short-circuit test	See clause 7.3.2.	P

7	SPECIFIC REQUIREMENTS AND TESTS		P
7.1	Charging procedure for test purposes		P
7.1.1	First procedure		P
	This charging procedure applies to subclauses other than those specified in 7.1.2		P
	Unless otherwise stated in this document, the charging procedure for test purposes is carried out in an ambient temperature of 20 °C $\pm$ 5 °C, using the method declared by the manufacturer		P
	Prior to charging, the battery has been discharged at 20 °C $\pm$ 5 °C at a constant current of 0,2 It A down to a specified final voltage		P
7.1.2	Second procedure		P
	This charging procedure applies only to 7.3.1, 7.3.4, 7.3.5, and 7.3.9		P

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Clause	Requirement + Test	Result - Remark	Verdict
	After stabilization for 1 h to 4 h, at an ambient temperature of the highest test temperature and the lowest test temperature, respectively, as specified in Table 2, cells are charged by using the upper limit charging voltage and maximum charging current, until the charging current is reduced to 0,05 It A, using a constant current to constant voltage charging method	Charge temperature range: 0°C - 45°C declared. -5°C used for lower limit tests. 45°C used for upper limit tests.	P
7.2	Intended use		P
7.2.1	Continuous charging at constant voltage (cells)		P
	Fully charged cells are subjected for 7 days to a charge using the charging method for current and standard voltage specified by the cell manufacturer	Charging for 7 days with 600mA.	P
	Results: no fire, no explosion, no leakage..... :	(See appended table 7.2.1)	P
7.2.2	Case stress at high ambient temperature (battery)	Tested complied.	P
	Oven temperature (°C)..... :	70°C	P
	Results: no physical distortion of the battery case resulting in exposure of internal protective components and cells	No physical distortion of the battery case resulting in exposure of internal protective components and cells.	P
7.3	Reasonably foreseeable misuse		P
7.3.1	External short-circuit (cell)	Tested complied.	P
	The cells were tested until one of the following occurred:		P
	- 24 hours elapsed; or		N/A
	- The case temperature declined by 20 % of the maximum temperature rise		P
	Results: no fire, no explosion..... :	(See appended table 7.3.1)	P
7.3.2	External short-circuit (battery)	Tested complied.	P
	The batteries were tested until one of the following occurred:		P
	- 24 hours elapsed; or		N/A
	- The case temperature declined by 20 % of the maximum temperature rise		P
	In case of rapid decline in short circuit current, the battery pack remained on test for an additional one hour after the current reached a low end steady state condition		P
	A single fault in the discharge protection circuit is conducted on one to four (depending upon the protection circuit) of the five samples before conducting the short-circuit test	Single fault conducted on three samples.	P

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Clause	Requirement + Test	Result - Remark	Verdict
	A single fault applies to protective component parts such as MOSFET (metal oxide semiconductor field-effect transistor), fuse, thermostat or positive temperature coefficient (PTC) thermistor	Single fault applies on MOSFET.	P
	Results: no fire, no explosion..... :	(See appended table 7.3.2)	P
7.3.3	Free fall	Tested complied.	P
	Results: no fire, no explosion	No fire. No explosion	P
7.3.4	Thermal abuse (cells)	Tested complied.	P
	Oven temperature (°C)..... :	130°C	—
	Results: no fire, no explosion	No fire. No explosion	P
7.3.5	Crush (cells)	Tested complied.	P
	The crushing force was released upon:		P
	- The maximum force of 13 kN ± 0,78 kN has been applied; or		P
	- An abrupt voltage drop of one-third of the original voltage has been obtained		N/A
	Results: no fire, no explosion..... :	(See appended table 7.3.5)	P
7.3.6	Over-charging of battery	Tested complied.	P
	The supply voltage which is:		P
	- 1,4 times the upper limit charging voltage presented in Table A.1 (but not to exceed 6,0 V) for single cell/cell block batteries or		N/A
	- 1,2 times the upper limit charging voltage resented in Table A.1 per cell for series connected multi-cell batteries, and	25.5V applied.	P
	- Sufficient to maintain a current of 2,0 It A throughout the duration of the test or until the supply voltage is reached		P
	Test was continued until the temperature of the outer casing:		P
	- Reached steady state conditions (less than 10 °C change in 30-minute period); or		P
	- Returned to ambient		N/A
	Results: no fire, no explosion..... :	(See appended table 7.3.6)	P
7.3.7	Forced discharge (cells)	Tested complied.	P
	Discharge a single cell to the lower limit discharge voltage specified by the cell manufacturer		P
	The discharged cell is then subjected to a forced discharge at 1 It A to the negative value of the upper limit charging voltage		P

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Clause	Requirement + Test	Result - Remark	Verdict
	- The discharge voltage reaches the negative value of upper limit charging voltage within the testing duration. The voltage is maintained at the negative value of the upper limit charging voltage by reducing the current for the remainder of the testing duration		N/A
	- The discharge voltage does not reach the negative value of upper limit charging voltage within the testing duration. The test is terminated at the end of the testing duration		P
	Results: no fire, no explosion..... :	(See appended table 7.3.7)	P
7.3.8	Mechanical tests (batteries)		P
7.3.8.1	Vibration	Tested complied.	P
	Results: no fire, no explosion, no rupture, no leakage or venting. :	(See appended table 7.3.8.1)	P
7.3.8.2	Mechanical shock	Tested complied.	P
	Results: no leakage, no venting, no rupture, no explosion and no fire :	(See appended table 7.3.8.2)	P
7.3.9	Design evaluation – Forced internal short-circuit (cells)	Tested complied.	P
	The cells complied with national requirement for :	France, Japan, Republic of Korea, Switzerland	—
	The pressing was stopped upon:		P
	- A voltage drop of 50 mV has been detected; or		N/A
	- The pressing force of 800 N (cylindrical cells) or 400 N (prismatic cells) has been reached	800N for cylindrical cells.	P
	Results: no fire..... :	(See appended table 7.3.9)	P
8	INFORMATION FOR SAFETY		P
8.1	General		P
	Manufacturers of secondary cells provides information about current, voltage and temperature limits of their products	Information for safety mentioned in manufacturer's specifications.	P
	Manufacturers of batteries provides information regarding how to minimize and mitigate hazards to equipment manufacturers or end-users	Information for safety mentioned in manufacturer's specifications.	P
	Systems analyses are performed by device manufacturers to ensure that a particular battery design prevents hazards from occurring during use of a product		N/A
	As appropriate, any information relating to hazard avoidance resulting from a system analysis is provided to the end user		N/A
8.2	Small cell and battery safety information	Not small cells and batteries.	N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	The following warning language is to be provided with the information packaged with the small cells and batteries or equipment using them:		N/A
	- Keep small cells and batteries which are considered swallowable out of the reach of children		N/A
	- Swallowing may lead to burns, perforation of soft tissue, and death. Severe burns can occur within 2 h of ingestion		N/A
	- In case of ingestion of a cell or battery, seek medical assistance promptly		N/A
9	MARKING		P
9.1	Cell marking		N/A
	Cells are marked as specified in IEC 61960, except coin cells	The final product is battery.	N/A
	Coin cells whose external surface area is too small to accommodate the markings on the cells show the designation and polarity		N/A
	By agreement between the cell manufacturer and the battery and/or end product manufacturer, component cells used in the manufacture of a battery need not be marked		N/A
9.2	Battery marking		P
	Batteries are marked as specified in IEC 61960, except for coin batteries	The battery is marked in accordance with IEC 61960, also see copy of marking plate.	P
	Coin batteries whose external surface area is too small to accommodate the markings on the batteries show the designation and polarity	Not coin batteries.	N/A
	Batteries are marked with an appropriate caution statement		P
	- Terminals have clear polarity marking on the external surface of the battery, or	The "+" and "-" polarity explicitly marked on surface of the battery.	P
	- Not be marked with polarity markings if the design of the external connector prevents reverse polarity connections		N/A
9.3	Caution for ingestion of small cells and batteries	Not small cells and batteries.	N/A
	Coin cells and batteries identified as small batteries include a caution statement regarding the hazards of ingestion in accordance with 8.2		N/A
	Small cells and batteries are intended for direct sale in consumer-replaceable applications, caution for ingestion is given on the immediate package		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
9.4	Other information		P
	The following information are marked on or supplied with the battery:		P
	- Storage and disposal instructions	Information for storage and disposal instructions mentioned in manufacturer's specifications.	P
	- Recommended charging instructions	Information for recommended charging instructions mentioned in manufacturer's specifications.	P

10	PACKAGING AND TRANSPORT		P
	Packaging for coin cells are not be small enough to fit within the limits of the ingestion gauge of Figure 3	Not coin cells.	N/A

ANNEX A	CHARGING AND DISCHARGING RANGE OF SECONDARY LITHIUM ION CELLS FOR SAFE USE		P
A.1	General		P
A.2	Safety of lithium ion secondary battery	Complied.	P
A.3	Consideration on charging voltage	Complied.	P
A.3.1	General		P
A.3.2	Upper limit charging voltage	4.25V applied.	P
A.3.2.1	General		P
A.3.2.2	Explanation of safety viewpoint		N/A
A.3.2.3	Safety requirements, when different upper limit charging voltage is applied	4.25V applied.	N/A
A.4	Consideration of temperature and charging current		P
A.4.1	General		P
A.4.2	Recommended temperature range	See A.4.2.2.	P
A.4.2.1	General		P
A.4.2.2	Safety consideration when a different recommended temperature range is applied	Charging temperature range declared by client is: 0-45°C	P
A.4.3	High temperature range	Charging high temperature declared by client is: 45°C.	P
A.4.3.1	General		N/A
A.4.3.2	Explanation of safety viewpoint		N/A
A.4.3.3	Safety considerations when specifying charging conditions in the high temperature range		N/A
A.4.3.4	Safety considerations when specifying a new upper limit in the high temperature range		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
A.4.4	Low temperature range	Charging low temperature declared by client is: 0°C.	P
A.4.4.1	General		P
A.4.4.2	Explanation of safety viewpoint		P
A.4.4.3	Safety considerations, when specifying charging conditions in the low temperature range		P
A.4.4.4	Safety considerations when specifying a new lower limit in the low temperature range	No documents provided by manufacturer explaining the lower limit exceed 10°C, -5°C applied for testing in this report for safety considerations.	P
A.4.5	Scope of the application of charging current		P
A.4.6	Consideration of discharge		P
A.4.6.1	General		P
A.4.6.2	Final discharge voltage and explanation of safety viewpoint	Cell specified final voltage 2.7V, not exceed 2.7V specified by cell manufacturer.	P
A.4.6.3	Discharge current and temperature range		P
A.4.6.4	Scope of application of the discharging current		P
A.5	Sample preparation		P
A.5.1	General		P
A.5.2	Insertion procedure for nickel particle to generate internal short		P
A.5.3	Disassembly of charged cell		P
A.5.4	Shape of nickel particle		P
A.5.5	Insertion of nickel particle in cylindrical cell		P
A.5.5.1	Insertion of nickel particle in winding core		P
A.5.5.2	Marking the position of the nickel particle on both ends of the winding core of the separator		P
A.5.6	Insertion of nickel particle in prismatic cell		N/A
A.6	Experimental procedure of the forced internal short-circuit test		P
A.6.1	Material and tools for preparation of nickel particle		P
A.6.2	Example of a nickel particle preparation procedure		P
A.6.3	Positioning (or placement) of a nickel particle		P
A.6.4	Damaged separator precaution		P
A.6.5	Caution for rewinding separator and electrode		P
A.6.6	Insulation film for preventing short-circuit		P
A.6.7	Caution when disassembling a cell		P

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Clause	Requirement + Test	Result - Remark	Verdict
A.6.8	Protective equipment for safety		P
A.6.9	Caution in the case of fire during disassembling		P
A.6.10	Caution for the disassembling process and pressing the electrode core		P
A.6.11	Recommended specifications for the pressing device		P
ANNEX B	RECOMMENDATIONS TO EQUIPMENT MANUFACTURERS AND BATTERY ASSEMBLERS		N/A
ANNEX C	RECOMMENDATIONS TO THE END-USERS		N/A
ANNEX D	MEASUREMENT OF THE INTERNAL AC RESISTANCE FOR COIN CELLS		N/A
D.1	General	Not coin cells.	N/A
D.2	Method		N/A
	A sample size of three coin cells is required for this measurement		N/A
	Coin cells with an internal resistance greater than 3 Ω require no further testing		N/A
	Coin cells with an internal resistance less than or equal to 3 Ω are subjected to the testing according to Clause 6 and Table 1		N/A
ANNEX E	PACKAGING AND TRANSPORT		N/A
ANNEX F	COMPONENT STANDARDS REFERENCES		N/A

5.1 – 5.6					
TABLE: Critical components information					
Object/part no.	Manufacturer/ trademark	Type/model	Technical data	Standard	Mark(s) of conformity
Plastic shell	FORMOSA CHEMICALS & FIBRE CORP PLASTICS DIV	AC310(+)	V-0, 1.5mm	UL94	UL E194560
PCB	Huizhou Huaxingbo Electronics Co Ltd	HXB-M1	V-0, 130°C, Min. thickness: 1.6mm	UL 94 UL 796	UL E524508
PCB (Alternative)	Interchangeable	Interchangeable	V-0, 130°C, Min. thickness: 1.6mm	UL 94 UL 796	UL approved
IC (U1)	Fremont Micro Devices Corporation	FT61F14X	V _{SS} : 2.1V ~ 5.5V, T _A : -40°C ~ 105°C	--	Tested with appliance
MOSFET (Q3, Q4)	Leshan Radio Company, LTD	L2N7002DW1 T1G	V _{DS} : 60V, V _{GS} : ±20V, I _D : ±115A	--	Tested with appliance
Cell	Shanxi Hengchangyuan Technology Co., LTD	INR18650P	3.6V, 3000mAh	IEC 62133- 2:2017/AM D1:2021	Tested with appliance
-Positive electrode	Xinxiang Tianli Lithium Energy Co., Ltd.	TLM510	LiN _{0.5} iCo _{0.2} Mn _{0.3} O ₂	--	--
-Negative electrode	Jiangxi zichen Technology Co.,Ltd	G56	Graphite, D50(um): 14±2um	--	--
-Separator	Jiangxi Yutu New Materials Co., Ltd	7+2+2	Thickness: 0.011mm PE+ Ceramic+ PVDF Shut Down Temperature(°C): 135	--	--
-Electrolyte	Guangdong Jinguang Hi Tech Co., Ltd	A2796	LiPF ₆ +EC+DEC Conductivity 25 °C: 7.5±0.5mS/cm	--	--
Supplementary information: N/A					

7.2.1	TABLE: Continuous charging at constant voltage (cells)				P
Sample no.	Recommended charging voltage Vc (Vdc)	Recommended charging current I _{rec} (A)	OCV before test (Vdc)	Results	
Cell #1	4.20	0.6	4.18	P	
Cell #2	4.20	0.6	4.19	P	
Cell #3	4.20	0.6	4.19	P	
Cell #4	4.20	0.6	4.18	P	
Cell #5	4.20	0.6	4.19	P	
Supplementary information: - No fire or explosion - No leakage					

7.3.1	TABLE: External short-circuit (cell)					P
Sample no.	Ambient T (°C)	OCV before test (Vdc)	Resistance of circuit (mΩ)	Maximum case temperature rise ΔT (°C)	Results	
Samples charged at charging temperature upper limit (45°C)						
Cell #6	55.7	4.22	85.3	102.0	P	
Cell #7	55.7	4.21	89.5	104.1	P	
Cell #8	55.7	4.22	83.7	105.3	P	
Cell #9	55.7	4.21	92.6	99.8	P	
Cell #10	55.7	4.21	85.4	102.1	P	
Samples charged at charging temperature lower limit (-5°C)						
Cell #11	55.9	4.17	90.4	102.4	P	
Cell #12	55.9	4.18	89.6	98.9	P	
Cell #13	55.9	4.17	83.4	97.8	P	
Cell #14	55.9	4.17	86.2	101.0	P	
Cell #15	55.9	4.16	88.6	96.6	P	
Supplementary information:						
- No fire or explosion						

7.3.2	TABLE: External short-circuit (battery)					P
Sample no.	Ambient T (°C)	OCV before test (Vdc)	Resistance of circuit (mΩ)	Maximum case temperature rise ΔT (°C)	Component single fault condition	Results
Battery #4	22.5	20.91	83.7	92.6	MOS	P
Battery #5	22.5	20.90	82.3	90.1	MOS	P
Battery #6	22.5	20.92	85.9	95.3	MOS	P
Battery #7	22.5	20.92	83.2	23.4	/	P
Battery #8	22.5	20.93	83.2	23.7	/	P
Supplementary information: - No fire or explosion						

7.3.5	TABLE: Crush (cells)				P
Sample no.	OCV before test (Vdc)	OCV at removal of crushing force (Vdc)	Maximum force applied to the cell during crush (kN)	Results	
Samples charged at charging temperature upper limit (45°C)					
Cell #29	4.21	4.20	13.00	P	
Cell #30	4.22	4.21	13.01	P	
Cell #31	4.22	4.22	13.02	P	
Cell #32	4.20	4.19	13.03	P	
Cell #33	4.21	4.21	13.05	P	
Samples charged at charging temperature lower limit (-5°C)					
Cell #34	4.17	4.17	13.02	P	
Cell #35	4.18	4.17	13.03	P	
Cell #36	4.16	4.16	13.07	P	
Cell #37	4.17	4.16	13.04	P	
Cell #38	4.18	4.17	13.04	P	
Supplementary information:					
- No fire or explosion					

7.3.6		TABLE: Over-charging of battery			P
Constant charging current (A)		12.0			—
Supply voltage (Vdc)		25.5			—
Sample no.	OCV before charging (Vdc)	Total charging time (minute)	Maximum outer case temperature (°C)	Results	
Battery #12	15.21	90	34.2	P	
Battery #13	15.24	90	33.2	P	
Battery #14	15.23	90	33.1	P	
Battery #15	15.25	90	34.3	P	
Battery #16	15.22	90	35.5	P	
Supplementary information:					
- No fire or explosion					

7.3.7	TABLE: Forced discharge (cells)				P
Sample no.	OCV before application of reverse charge (Vdc)	Measured reverse charge I_t (A)	Lower limit discharge voltage (Vdc)	Results	
Cell #39	3.06	3.0	2.7	P	
Cell #40	3.07	3.0	2.7	P	
Cell #41	3.06	3.0	2.7	P	
Cell #42	3.05	3.0	2.7	P	
Cell #43	3.04	3.0	2.7	P	
Supplementary information:					
- No fire or explosion					

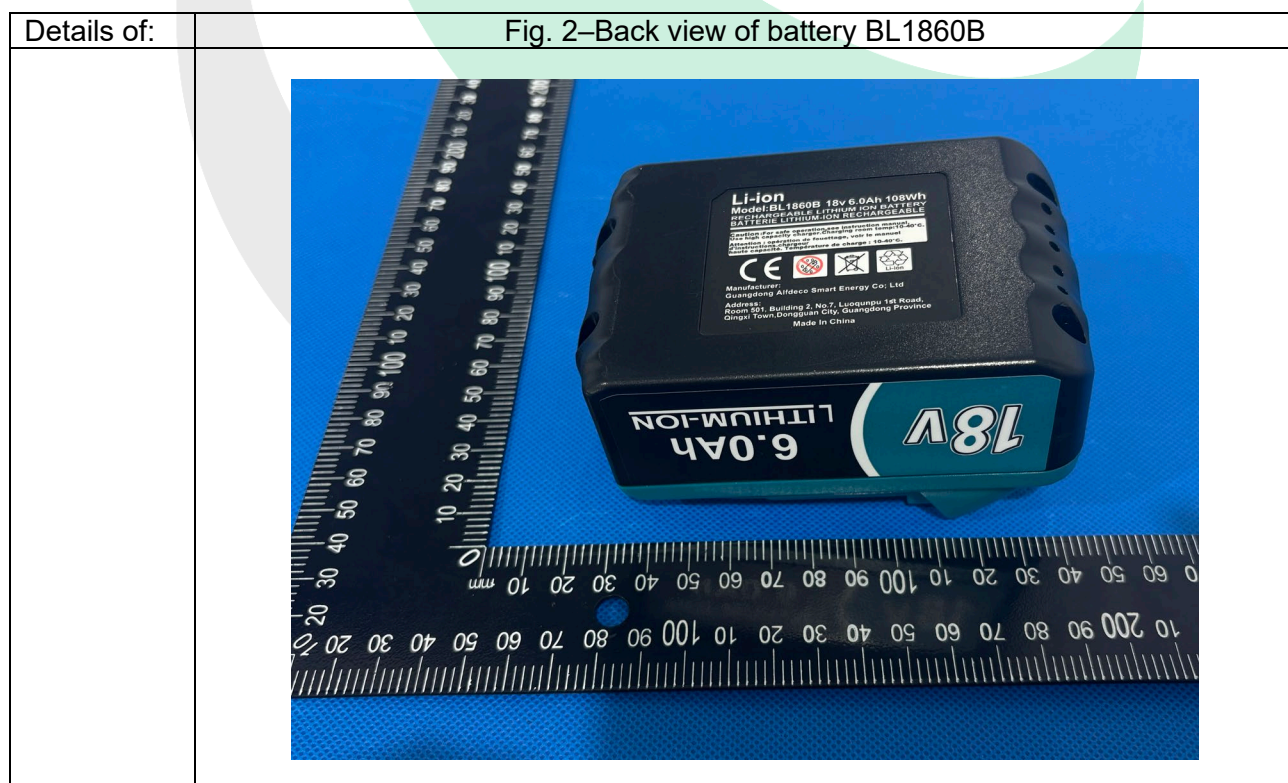
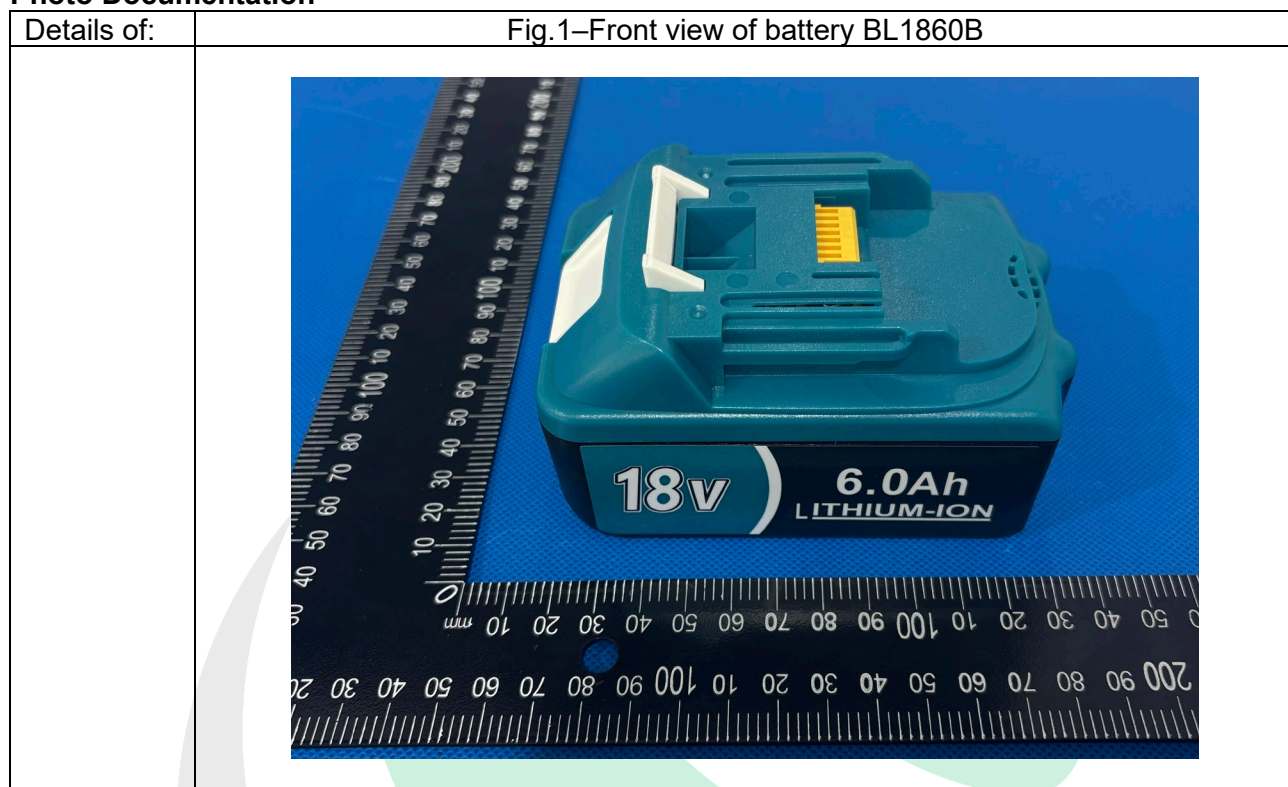
7.3.8.1	TABLE: Vibration				P
Sample no.	OCV before test (Vdc)	OCV after test (Vdc)	Mass before test (g)	Mass after test(g)	Results
Battery #17	20.91	20.89	570.54	570.51	P
Battery #18	20.91	20.91	570.48	570.42	P
Battery #19	20.90	20.87	570.51	570.44	P
Supplementary information: - No fire or explosion - No rupture - No leakage - No venting					

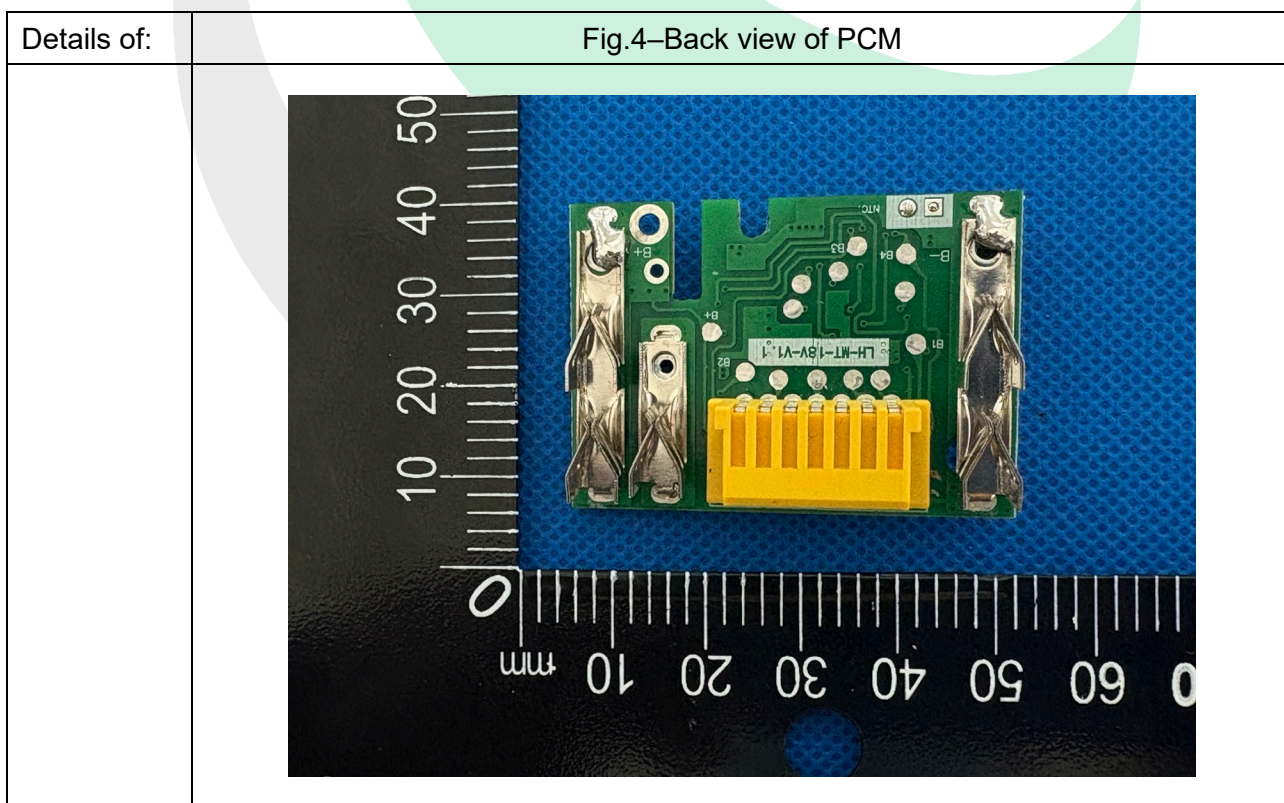
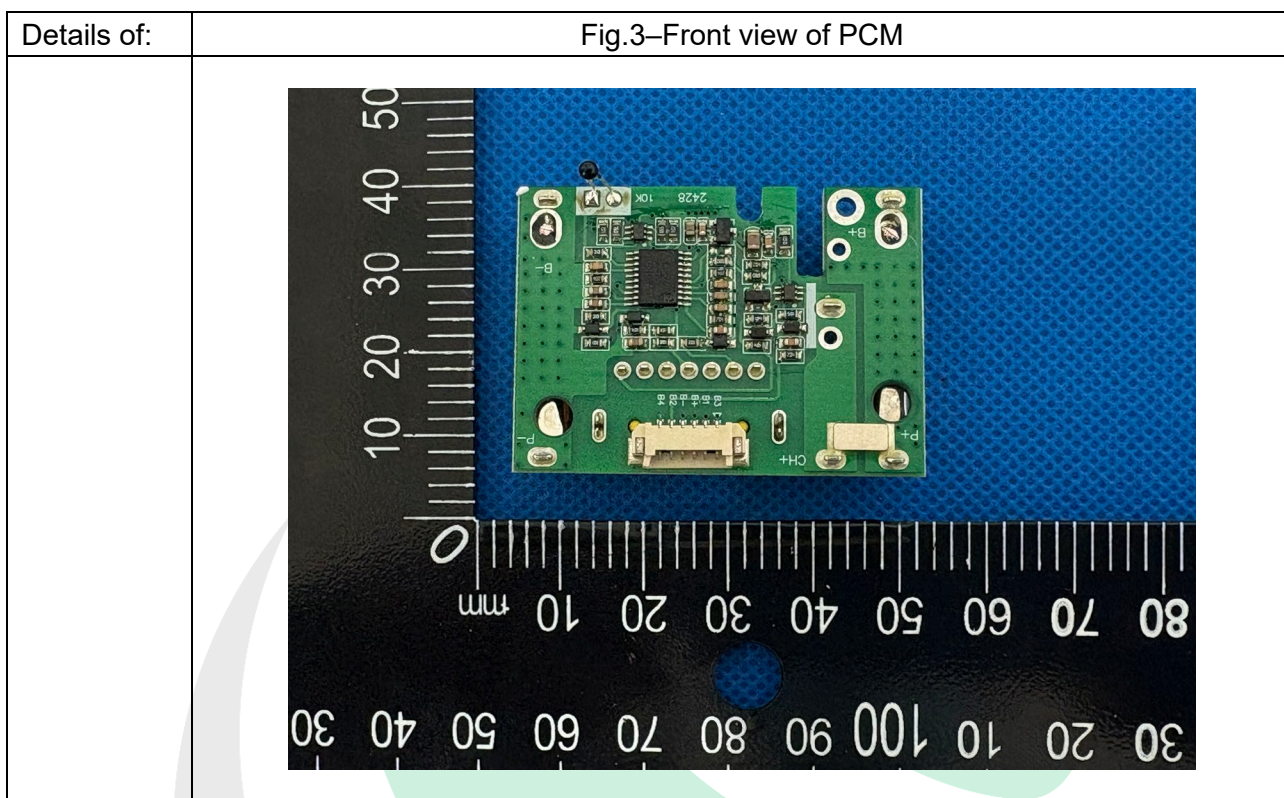
7.3.8.2	TABLE: Mechanical shock					P
Sample no.	OCV before test (Vdc)	OCV after test (Vdc)	Mass before test (g)	Mass after test(g)	Results	
Battery #20	20.92	20.92	570.53	570.51	P	
Battery #21	20.90	20.90	570.52	570.51	P	
Battery #22	20.93	20.91	570.49	570.49	P	
Supplementary information: - No fire or explosion - No rupture - No leakage - No venting						

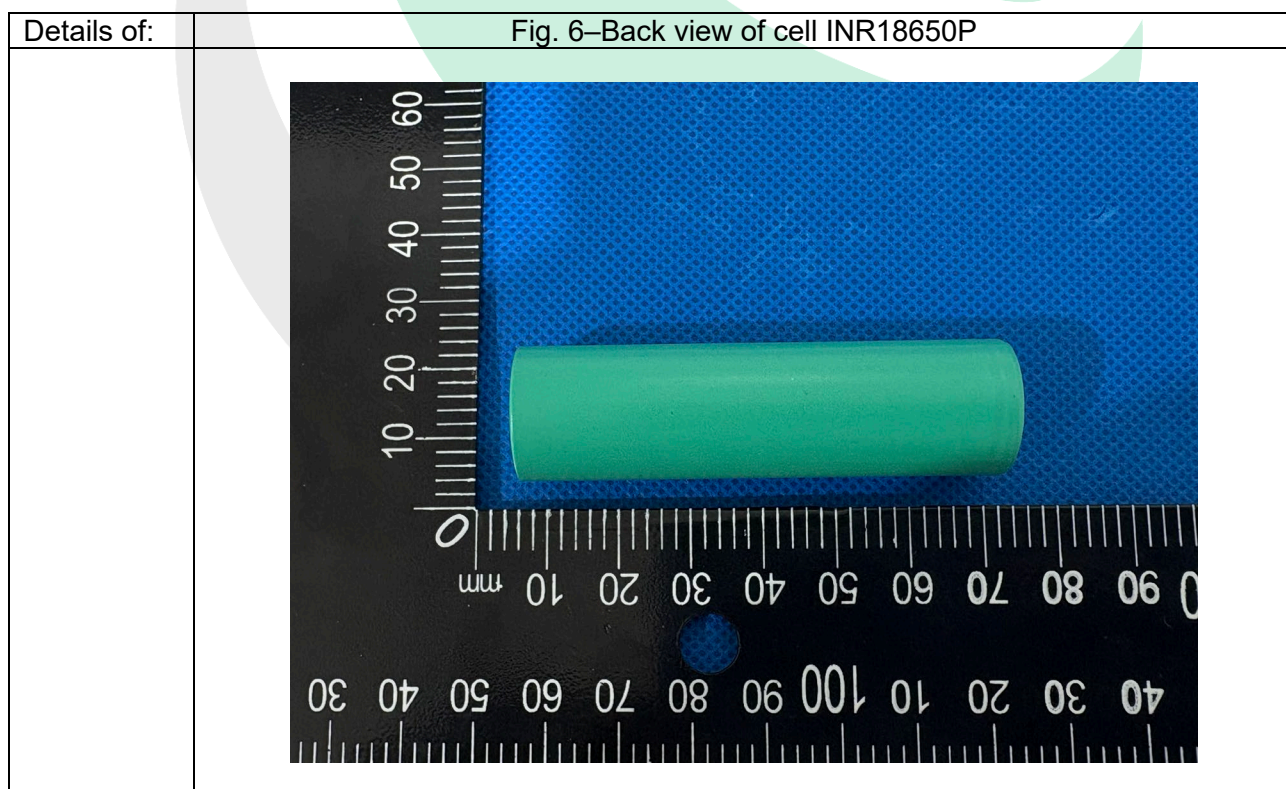
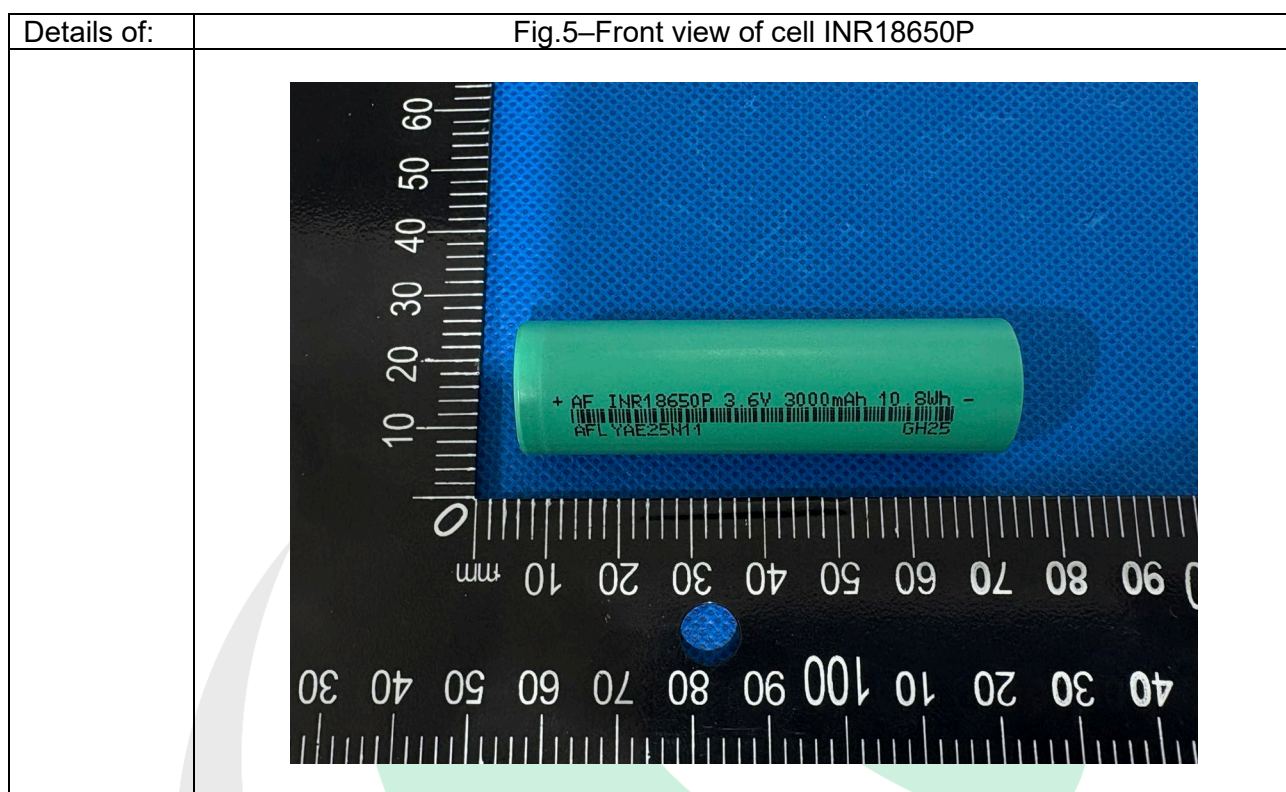
7.3.9	TABLE: Forced internal short circuit (cells)					P
Sample no.	Chamber ambient T (°C)	OCV before test (Vdc)	Particle location ¹⁾	Maximum applied pressure (N)	Results	
Samples charged at charging temperature upper limit (45°C)						
Cell #44	45	4.22	1	800	P	
Cell #45	45	4.21	1	800	P	
Cell #46	45	4.21	1	800	P	
Cell #47	45	4.22	1	800	P	
Cell #48	45	4.21	1	800	P	
Samples charged at charging temperature lower limit (-5°C)						
Cell #49	-5	4.18	1	800	P	
Cell #50	-5	4.17	1	800	P	
Cell #51	-5	4.17	1	800	P	
Cell #52	-5	4.16	1	800	P	
Cell #53	-5	4.17	1	800	P	
Supplementary information: ¹⁾ Identify one of the following: 1: Nickel particle inserted between positive and negative (active material) coated area. 2: Nickel particle inserted between positive aluminium foil and negative active material coated area. - No fire or explosion						

D.2	TABLE: Internal AC resistance for coin cells				N/A
Sample no.	Ambient T (°C)	Store time (h)	Resistance Rac (Ω)	Results ¹⁾	
Supplementary information: ¹⁾ Coin cells with internal resistance less than or equal to 3 Ω, see test result on corresponding tables					

Appendix 1
Photo Documentation







---End of Test Report---